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A comparative study of TVD-limiters—well-known limiters and an introduction of new ones

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SUMMARY

This paper gives a comparative study of TVD-limiters for standard explicit Finite Volume schemes. In contrast to older studies, it includes also unsymmetrical limiter functions that depend on the local CFL-number. We classify the limiters and show how to extend these families of limiters. We introduce a new member of the Superbee family, which is adapted to Roe's linear third-order scheme. Based on an idea by Serna and Marquina, new smooth limiters are introduced, which turn the van Leer and van Albada limiters into complete classes of limiters. The comparison of the limiters is done with some standard test cases. The results clarify the influence of the chosen limiter on the quality of the numerical results. Compared with ENO or WENO schemes, they also show the high resolution, which can be obtained by a CFL-number-dependent limiter when the grid is not highly refined. Copyright © 2010 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Although with ENO and WENO schemes there are higher-order alternatives, TVD-schemes‡ are still in widespread use. One reason for that is the simple implementation. Particularly on irregular meshes, it is not always easy to extend high-order methods, such as ENO and WENO, to two or three space-dimensions. Other reasons are the—compared with ENO and WENO—compact stencil and the robustness of the resulting schemes. Unfortunately, in the literature only a few limiters are treated in more detail. Many limiters are only found or cited in one or two papers. Some of them are just given as an incidental remark. As a result, only a small class of limiters is found in textbooks and therefore in practical use. This class is restricted in two senses: it only contains limiters that do not take into account the local CFL number, and it only contains symmetric limiter functions, i.e. the information on the upwind direction is not used. As is pointed out by Roe [2], this is mainly due to the fact that in the 1980s most simulations were done with implicit codes for steady states. In this case, the choice of the limiter function is restricted to the above-mentioned class and without too much impact on the quality of the results. For explicit codes, there is much more

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‡The term TVD-scheme, although standard in literature, is somehow misleading. For several space-dimensions, total variation diminishing would be a severe restriction [1], for non-linear systems, it would be nonsense as can be seen, for example, from the Shu–Osher test problem in Section 5.2. The term only refers to the fact that a restriction of the scheme to 1d linear advection would be TVD.

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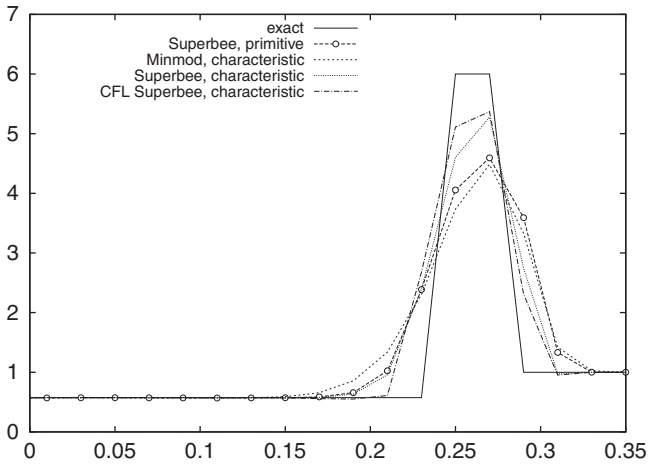


Figure 1. Toro's third test problem with different limiting strategies. The picture shows a density plot of the most interesting region.

freedom in the choice of limiters. It is possible to use the CFL-dependent limiters introduced by van Leer [3] and Roe and coworkers [2, 4–7]. Another reason for the restriction to CFL-independent limiters is the widespread practise to reconstruct and limit in primitive rather than in characteristic variables. Only for characteristic variables, it is possible to take into account the CFL-number for a specific wave. As Figure 1 shows, it is disadvantageous to reconstruct in primitive variables. The Superbee in primitive variables is only slightly better than the Minmod limiter in characteristic variables. Applying Superbee to characteristic variables leads to a significant improvement. A further improvement can be obtained by using its CFL-dependent counterpart.[§]

Very high-order schemes, such as ENO and WENO, show their nice performance on very fine grids. But, especially in 3d, due to the high computational effort, one often has to resort to coarser grids. On coarse grids the performance of such a scheme is, due to the high order, rather poor. It turns out that on many practical relevant grids, the resolution of a suitable TVD-scheme is comparable to the resolution of ENO- or WENO-schemes while the computational cost is much lower.

In the last years, some comparative studies of numerical schemes were published, e.g. [9, 10], which compare different types of schemes, e.g. TVD-schemes to WENO schemes. Unfortunately all of them treat the schemes as a fixed one, with a fixed setting of parameters and a fixed limiter. Therefore, one cannot find any information on the influence of the limiter used.[¶] It turns out that in some cases the results for the TVD-scheme could be considerably improved by the choice of an alternative limiter. Waterson and Deconinck [12] compare limiters designed for the use in semi-discrete schemes. They consider a greater variety of limiters than Sweby in his famous paper [11], but still the class of limiters is restricted by excluding any reference to the upwind direction or even the local CFL-number. The goal of this study is to fill in that gap and provide users of TVD-schemes with a guidance how to choose an appropriate limiter for a given application. For this purpose, we first divide the limiters in classes or families, which contain limiters based on the same construction principles. Almost all of these classes allow for further generalizations or enhancements. For the Superbee family, we present a new third-order adapted variant, which turns out to be the best candidate for an all purpose limiter.

[§]Roe and Baines [6] called this counterpart Ultrabee. Since the term is used by Toro [8] for another function, we call it CFL-Superbee in this paper.

[¶]The famous paper of Sweby [11] compares only a small number of limiters.

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In the course of the development of their WENO scheme, Serna and Marquina [13] give a generalization of the harmonic mean, which also includes the well known minmod limiter as a special case. By further generalizations this technique can be used to create new limiters which take into account the local CFL-number and share with the van Leer limiter the smooth behaviour in monotone regions. Due to its smooth behaviour, it turns out to be the best choice for compressive non-linear waves. By adopting the core idea of Serna and Marquina, writing the limiter as a multiplicative correction of the arithmetic mean, we also construct enhanced variants of the van Albada limiter. The new class also contains an improved variant which is applicable without any knowledge of the local CFL-number.

In the last decade a number of limiters were introduced, which are not TVD, but still keep the total variation bounded in some manner. Among them are the piecewise hyperbolic [14] and the double logarithmic limiter [15]. It is originally designed for the exclusive use in semi-discrete schemes. Therefore, it does not take into account the local CFL-number. For comparison we include one of these modern limiters in this study. We have chosen a limiter by Čada and Torrilhon [16], which shares the main properties with the double logarithmic limiter. We present a modification of this limiter, which allows to adopt it to the local CFL-number. This leads to a significant improvement in the numerical results, especially for small CFL-numbers (slowly moving entropy waves etc.).

The paper is organized as follows: First we give a review of the basic theory by van Leer, Harten and Roe. Then we present the different classes of limiters and show their nature by the results on a linear advection problem, followed by a convergence study. For this we use the advection test problem, which includes kinks and discontinuities and therefore serves as a good measure for the overall behaviour of the resulting schemes, and a simple sinusoidal wave-packet. We also present results with common test cases for the 1d and 2d Euler-equations.

Except for the computations with limiting in primitive variables presented in Figure 1, all numerical computations are done with *clawpack* [17]. This is an implementation of the wave propagation approach, described in much detail in [18]. The limiters are applied to waves arising from solutions to the Riemann problems at the cell faces. Thus, it is an algebraic limiting, no geometric limiting. Especially for non-linear systems and in 2d, the results might differ from those obtained with other implementations.

2. REVIEW OF THE BASIC THEORY

In this section we review the basic theory for TVD-schemes. The starting point is van Leer's REA algorithm [3]. Considering this and Roe's theory of linear schemes [4] for the advection equation, it is possible to find criteria for the limiting of the total variation.

2.1. The REA algorithm

In his fundamental work [3] van Leer introduces the REA-algorithm, Reconstruct–Evolve–Average, for the linear advection equation

$$q_t + aq_x = 0. \tag{1}$$

We consider an equidistant grid with cell size Δx and time step Δt. The grid cells are $[x_{i-1/2}, x_{i+1/2}]$ with cell midpoint x_i. We assume the velocity a to be positive. Thus, the CFL-number $\nu = a\Delta t/\Delta x$ is also positive. Owing to stability reasons, we only consider CFL-numbers $\nu \leqslant 1$. For van Leer's scheme I, which is the standard for TVD-schemes nowadays, at time t^n in each cell only the cell average q_i^n is given. If we reconstruct $q(x, t^n)$ with a piecewise linear function

$$\hat{Q}(x, t^n) = q_i^n + s_i^n (x - x_i) \quad \text{for } x \in [x_{i-1/2}, x_{i+1/2}] \tag{2}$$

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and evolve according to advection equation (1) up to time t^{n+1}, then, after averaging in the cells, we get

$$q_i^{n+1} = q_i^n - v(q_i^n - q_{i-1}^n) - \tfrac{1}{2}\Delta x\, v(1-v)(s_i^n - s_{i-1}^n). \tag{3}$$

The question now is how to determine the slopes s_i. A natural choice is a mean $s_i = \mathscr{M}(\Delta q_{i-1/2}/\Delta x, \Delta q_{i+1/2}/\Delta x)$ of the upwind and the downwind differences.

For general speeds a, in (3) the index $i-1$ has to be replaced by the upwind index, v has to be replaced by $|v|$, and the slope s_i has to be determined as

$$s_i = \mathscr{M}\left(\frac{\Delta q_{\text{up}}}{\Delta x}, \frac{\Delta q_{\text{down}}}{\Delta x}\right), \tag{4}$$

where the index up denotes the upwind difference and the index down the downwind difference.

The problem is now reduced to the choice of the mean $\mathscr{M}$. Therefore, as a next step, we have to look for conditions that a reasonable mean function should satisfy.

2.1.1. Properties of mean functions. A mean of two numbers usually is characterized by the following conditions.

Consistency:

$$\mathscr{M}(a,a) = a. \tag{M1}$$

Inclusion:

$$\min\{a,b\} \leqslant \mathscr{M}(a,b) \leqslant \max\{a,b\}. \tag{M2}$$

This is a generalization of the consistency (M1). It is not completely clear if this is necessary for the construction of a useful limiter. But if it is abandoned, the mean function should at least be consistent and Lipschitz continuous to get a reasonable numerical scheme.

Homogeneity:

$$\mathscr{M}(\lambda a, \lambda b) = \lambda \mathscr{M}(a,b). \tag{M3}$$

This is crucial since a numerical scheme should not depend on the reference values used to get the non-dimensional form of the physical system.

Symmetry:

$$\mathscr{M}(a,b) = \mathscr{M}(b,a). \tag{M4}$$

This is crucial for limiters that do not depend on the local CFL-number and thus on the strength of the upwinding. But it might be given up for limiters that take into account the CFL-number. As will be explained in Section 2.4.2, it might also be given up in semi-discrete schemes when, at least, the local upwind direction is known.

Monotonicity:

$$\mathscr{M}(\cdot, b) \quad \text{and} \quad \mathscr{M}(a, \cdot) \quad \text{are non-decreasing.} \tag{M5}$$

Since homogeneity is crucial for a mean function, we can introduce a normal form

$$\varphi_{\mathscr{M}}(r) = \mathscr{M}(r,1). \tag{5}$$

As a consequence, we find for non-vanishing b

$$\mathscr{M}(a,b) = b\mathscr{M}\left(\frac{a}{b}, 1\right) = b\varphi_{\mathscr{M}}\left(\frac{a}{b}\right) \tag{6}$$

and similarly for non-vanishing a

$$\mathscr{M}(b,a) = a\varphi_{\mathscr{M}}\left(\frac{b}{a}\right). \tag{7}$$

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With these equations, it is easy to write the main properties for mean functions in terms of the normal form:

$$\varphi(1)=1, \tag{M1$'$}$$

$$\min\{r,1\}\leqslant\varphi(r)\leqslant\max\{r,1\}, \tag{M2$'$}$$

$$\varphi\left(\frac{1}{r}\right)=\frac{1}{r}\varphi(r), \tag{M4$'$}$$

$$\varphi(r)\quad\text{and}\quad r\,\varphi\left(\frac{1}{r}\right)\quad\text{non-decreasing.} \tag{M5$'$}$$

For the monotonicity of symmetric means, it is sufficient that $\varphi(r)$ is non-decreasing.

2.1.2. Application to REA-algorithm. It is possible to rewrite the REA-algorithm in terms of the normal form of a mean function by setting

$$r=\frac{\Delta q_{\text{up}}}{\Delta q_{\text{down}}}. \tag{8}$$

From this, we get for the above example with positive signal speed $r_i=\Delta q_{i-1/2}/\Delta q_{i+1/2}$. Thus, the scheme (3) with slopes (4) reads as

$$q_i^{n+1}=q_i^n-\left[v+\frac{1}{2}v(1-v)\left(\frac{\varphi(r_i^n)}{r_i^n}-\varphi(r_{i-1}^n)\right)\right]\Delta q_{i-1/2}^n. \tag{9}$$

For $\varphi(r)\equiv 1$ this is just the Lax–Wendroff scheme, a 3-point central scheme, for $\varphi(r)=r$ the Beam–Warming scheme, which is based on a full upwind 3-point stencil, and for $\varphi(r)=(1+r)/2$ the Fromm scheme, which as the arithmetic mean of both is based on an upwind biased 4-point stencil. Obviously, all schemes of form (9) are based on the same stencil as the Fromm scheme. Since the paper restricts itself to schemes of this form, all schemes in our study are restricted to the stencil of the Fromm scheme. If we choose $\varphi(r)$ to be a linear function, then the resulting scheme is also linear.

2.2. Theory for linear schemes

In his fundamental paper [4], Roe comes up with a basic theory for constant coefficient (linear) schemes. It turns out that this gives a deep insight into the behaviour of TVD-schemes and limiters. Here we cite the main results.

First, it is obvious that any linear combination of pth order schemes is also of pth order. Therefore, the question arises if we can find a basis for the schemes of a given order and how much basic schemes we need.

Theorem 1
The linear schemes of order p with m-point support form a family describable by $m-p-1$ parameters.

Since the REA-algorithm with the above setting has a 4-point support, the schemes of second-order form a one-parameter family. All of these schemes can be written as a weighted mean of the Lax–Wendroff and the Beam–Warming scheme. There is even a (unique) third-order scheme in this family:

Theorem 2
The scheme $(1-(1+v)/3)\text{LW}+[(1+v)/3]\text{BW}$ is of third order.

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The weight given to the upwind scheme (Beam Warming) increases with the CFL-number. As a direct consequence, the mean function with normal form

$$\varphi_3(r)=1-\frac{1+|v|}{3}+\frac{1+|v|}{3}r=1+\frac{1+|v|}{3}(r-1) \tag{10}$$

results in a third-order scheme. Apparently, this mean function is not symmetric. Therefore, it seems to be reasonable to sacrifice symmetry (M4) in order to get a better resolution.

As Arora and Roe state in [2], any scheme with

$$\varphi(1+\varepsilon)=1+\mathcal{O}(\varepsilon) \tag{11}$$

is of second order.

Since the transformation to primitive or conservative variables is sufficiently smooth, even schemes with reconstruction in primitive or conservative variables show up second-order behaviour. Condition (11) directly transfers from one set of variables to the others.

Another result of Roe [4] is the following:

Theorem 3
If the initial condition for the discrete advection equation is a single Fourier component, then a linear scheme of even order p approximates the phase with order p and the amplitude with order $p+2$. If p is odd, then both, phase and amplitude, are approximated with order $p+1$.

Therefore, the third-order scheme given by Theorem 2 approximates phase and amplitude with fourth order. The second-order schemes approximate the phase with second order and the amplitude with fourth order. The effects of this can be seen in Figure 2. Even for the relatively well-behaved Fromm scheme, the error in the phase is significantly higher than for the third-order scheme. Since the initial state for Figure 2, which in this case is the exact solution again, might be considered as a superposition of plane waves (Fourier expansion), the lowered dispersion error for the third-order scheme also leads to a better approximation of the shape of the waves. For limiters that start from the third-order scheme we, therefore, expect a better representation of the shape of the solution. On the other hand, for schemes that are optimized for the amplitude we might lose this behaviour. In Section 3.1.2, we present a way to bridge this contradiction by our new $\beta=\frac{2}{3}$ limiter.

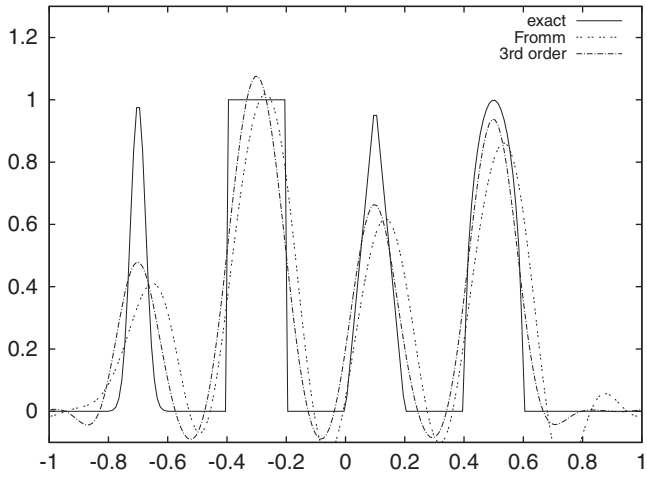


Figure 2. Application of linear schemes to the Zalesak test with CFL-number 0.1 ($t=20$).

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2.3. Bounding the total variation

In one space-dimension, the total variation of a discrete state can be written as

$$\mathrm{TV}(Q^n)=\sum_{i\in\mathbb{Z}}|\Delta q^n_{i+1/2}|. \tag{12}$$

A scheme is called TVD if the total variation does not grow in time. Harten [19] gives the following sufficient condition:

Theorem 4
Let a numerical scheme for a scalar hyperbolic conservation law be written in the form

$$q_i^{n+1}=q_i^n-C^n_{i-1/2}\Delta q^n_{i-1/2}+D^n_{i+1/2}\Delta q^n_{i+1/2}. \tag{13}$$

If for all $i\in\mathbb{Z}$ and all non-negative integers n, the coefficients satisfy the following two conditions:

$$C^n_{i+1/2},\, D^n_{i+1/2}\geqslant 0, \tag{14}$$

$$C^n_{i+1/2}+D^n_{i+1/2}\leqslant 1, \tag{15}$$

then the scheme is TVD.

For van Leer's scheme I in the form (9), the coefficients $D_{i+1/2}$ vanish. To satisfy conditions (14) and (15), the other coefficients have to satisfy

$$0\leqslant C^n_{i+1/2}=\nu+\frac{1}{2}\nu(1-\nu)\left(\frac{\varphi(r_i^n)}{r_i^n}-\varphi(r^n_{i-1})\right)\leqslant 1, \tag{16}$$

which is equivalent to

$$-\frac{2}{1-\nu}\leqslant\frac{\varphi(r_i^n)}{r_i^n}-\varphi(r^n_{i-1})\leqslant\frac{2}{\nu},$$

and for arbitrary upwind directions, we get the *General TVD-condition*:

$$-\frac{2}{1-|\nu|}\leqslant\frac{\varphi(r)}{r}-\varphi(R)\leqslant\frac{2}{|\nu|}\quad\forall r,\, R\in\mathbb{R}. \tag{17}$$

In some cases, it is necessary to have a mean function φ for which already the reconstruction step is TVD, e.g. to get a semi-discrete form and then apply a strong stability preserving (SSP) time scheme. These cases include implicit schemes as well as some schemes for non-linear transport. We have to consider two limiting cases:

1. no transport, i.e. $\nu\to 0$,
2. transport by exactly one grid cell, i.e. $|\nu|\to 1$.

A mean function that is TVD for both limiting cases, therefore, has to satisfy the *Restricted TVD-condition*

$$-2\leqslant\frac{\varphi(r)}{r}-\varphi(R)\leqslant 2\quad\forall r,\, R\in\mathbb{R}. \tag{18}$$

This is the form of the TVD-condition that is usually given in the literature. It is a necessary condition for the reconstruction step to be TVD. To the knowledge of the author, all limiters explicitly given in textbooks, except one example in [8], namely the upper-bound limiter (22), are constructed to satisfy this restricted TVD-condition. Besides the focus on implicit schemes, as mentioned in the introduction, this might be another important reason why there is only scarce use of limiters that take into account the local CFL-number and, thus, use the whole freedom given by the General TVD-condition (17).

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In order to make the reconstruction step TVD, it also has to be avoided that extrema increase. They should be reconstructed by a constant state, i.e. with zero slope. As a consequence, φ has to satisfy

$$\varphi(r)=0 \quad \text{for } r\leqslant 0. \tag{19}$$

Most limiters satisfy this condition. But in practice, violating condition (19) causes no problems, not even in the simulation of non-linear systems like gas dynamics.

If we set $\varphi(r)=0$ or $\varphi(R)=0$ in the General TVD-condition (17), we get the two conditions

$$\varphi(R)\leqslant \frac{2}{1-|v|}, \tag{20}$$

$$\varphi(r)\leqslant \frac{2}{|v|}r. \tag{21}$$

Together with (19) these are sufficient for a scheme to be TVD. We call the limiter function φ_{UB} with

$$\varphi_{\mathrm{UB}}(r)=\max\left\{0,\min\left\{\frac{2}{|v|}r,\frac{2}{1-|v|}\right\}\right\} \tag{22}$$

the upper bound for the limiters. This will play an important role in the construction of high-quality limiters. Toro [8] refers to this as Ultrabee. This might lead to confusion, since Roe and Baines [6] used this name for a special second-order limiter.

Another mean function that we will frequently use as a building block in the construction of limiters is the function φ_3, as given in (10), which defines the third-order scheme. If the limiter is to be used in a semi-discrete scheme, the local CFL-number is not available. Artebrandt and Schroll [15] found that in this case, $\varphi_3(r)$ has to be replaced by

$$\varphi_{3,0}(r)=1+\tfrac{1}{3}(r-1), \tag{23}$$

which is nothing else than (5) evaluated for CFL-number $v=0$.

In the same manner as (21) and (20) one obtains the conditions

$$\varphi(r)\geqslant -\frac{2}{1-|v|}r, \tag{24}$$

$$\varphi(R)\geqslant -\frac{2}{|v|}. \tag{25}$$

From (21) and (24) together, we conclude that any TVD limiter has to satisfy

$$\varphi(0)=0;. \tag{26}$$

Condition (25) turns out to be a useful tool for improving the (old) limiter by Čada *et al.* [16] in the end of the next section.

2.4. Construction principles for limiters

From the above results, we can deduce some general construction principles for limiters. We have to distinguish two cases: The case when the local CFL-number is available. This is the main focus of the paper. The second case is the use of the limiter in a semi-discrete scheme where the local CFL-number is not available and the reconstruction step itself has to be TVD.

2.4.1. When the local CFL-number is available. First we find two extremal cases, namely the limiter $\varphi(r)\equiv 0$ and the upper-bound φ_{UB}. Both do not satisfy condition (11); thus, both result in first-order schemes. The first one results in the standard upwind scheme, which is known to suffer from numerical viscosity, everything is smeared out. For the second, the loss of order results

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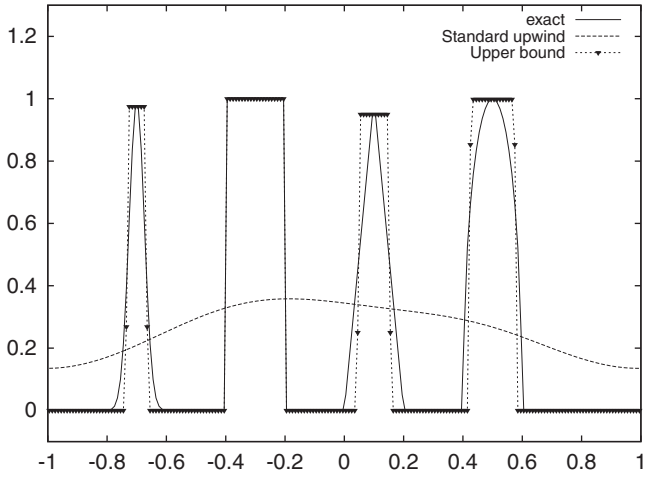


Figure 3. Zalesak test with first-order schemes at time $t=20$, the standard upwind and the compressive upper-bound scheme.

from a strong squaring effect. For a theoretical investigation of the second scheme, see [20]. As Figure 3 shows, for the standard upwind the complete structure of the solution is gone whereas for the upper-bound scheme the solution consists only of square waves. This squaring is unphysical for linear advection; it is only expected for non-linear compressive waves. Therefore, if one limiter tends more to smearing, whereas the other one tends more to squaring, we call the second one more compressive than the first one. The tuning of this artificial compression is essential for the quality of the numerical results.

A second issue is the connection to the third-order scheme. It seems to be preferable to stick to the third-order scheme as long as the TVD-condition (17) is not violated. This is possible for piecewise linear limiter functions. For smooth limiters, we can only expect the function to be tangent to the third-order scheme in $r=1$. As Roe points out [7], the condition

$$\varphi(1+\varepsilon)=\varphi_3(1+\varepsilon)+O(\varepsilon^2) \tag{27}$$

is sufficient for a scheme to show up third-order behaviour near $r=1$ and, thus, in smooth monotone regions of the solution.

2.4.2. Limiters for semi-discrete schemes. To use a limiter in a semi-discrete scheme, we require the reconstruction step itself to be TVD. Thus, we have to replace the General TVD-condition (17) by the restricted TVD-condition (18). As a consequence, in the construction of limiters, we have to replace the upper-bound φ_{UB} by the Sweby-bound

$$\varphi_{\mathrm{Sweby}}(r)=\max\{0,\min\{2r,2\}\}. \tag{28}$$

If the local upwind direction is known, φ_3 has to be replaced by $\varphi_{3,0}$ as mentioned above to make the reconstruction step, which corresponds to CFL-number $v=0$, third order. If the local upwind direction is not available, we should resort to a symmetric limiter function. A variety of such limiters can be found in [12]. Many of these limiters can be deduced from the limiters we present in the next section just by applying the replacements given in this paragraph.

2.5. The Zalesak test problem

In the following section, the properties of the limiters are illustrated by the numerical results for the test case of Zalesak [21], which was later on used by Jiang and Shu [22]. The only difference to the original setting is that we do the calculation for a longer time interval. The wave speed a in (1) is set to 1. Periodic boundary conditions are applied. Hence, analytically, after each full round, the state has to be the same as the initial state. The initial data contain a smooth but narrow

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combination of Gaussians, a square wave, a sharp triangle wave, and a half ellipse in the interval $[-1, 1]$. In detail,

$$q_0(x) = \begin{cases} \frac{1}{6}(G(x,\beta,z-\delta)+G(x,\beta,z+\delta)+4G(x,\beta,z)), & x\in[-0.8,-0.6]; \\ 1, & x\in[-0.4,-0.2]; \\ 1-|10(x-\frac{1}{10})|, & x\in[0,0.2]; \\ \frac{1}{6}(F(x,\alpha,a-\delta)+F(x,\alpha,a+\delta)+4F(x,\alpha,a)), & x\in[0.4,0.6]; \\ 0 & \text{otherwise} \end{cases} \tag{29}$$

with

$$G(x,\beta,z) = \mathrm{e}^{-\beta(x-z)^2},$$

$$F(x,\alpha,a) = \sqrt{\max\{1-\alpha^2(x-a)^2, 0\}},$$

and the parameters

$$a=0.5, \quad z=-0.7, \quad \delta=0.005, \qquad \alpha=10, \quad \beta=(\log 2)/(36\delta^2). \tag{30}$$

The calculations are done on a grid of 200 cells up to time 20, which corresponds to 10 full rounds. They are made with the CFL-numbers 0.1, 0.5, and 0.9. To keep the presentation clear, for some limiters we omit calculations that do not give additional information, most often the case with CFL-number 0.5.

3. THE LIMITERS

Since there are lots of limiters found in the literature and even more can be constructed by using the theory of the foregoing section, we have to classify them in some way. One important subclass is the class of piecewise linear mean functions. The well-known Monotoniced-centered (MC)-, Superbee-, and Minmod-limiters are members of this class. Another class contains limiter functions that are smooth at least apart of $r=0$. Famous members of this class are the van Leer-limiter (harmonic mean) and the van Albada-limiter. We will present the main ideas of both classes and show how some of the already existing limiters can be modified to enhance the quality of the simulation results. Besides these two classes, there are special techniques like shock switches. Here we will portray only the switch by Harten, because it is simple and its stencil is the same as for the limiters in consideration. In the end of this section, we present the Čada–Torrilhon limiter [16] as an example of a modern non-TVD limiter. For this limiter, we also show a modification that considerably improves the numerical results.

3.1. Piecewise linear limiter functions

First we consider piecewise linear limiter functions. They switch between different second-order schemes, or even the third-order scheme, depending on the local ratio r defined in Equation (8).

3.1.1. The MC-family. Already in [3] van Leer suggests to limit the Fromm scheme by the restricted TVD-condition (18) and condition (19), which prevents local extrema from being reconstructed. This results in the so-called MC-limiter

$$\varphi_{\mathrm{MC}}(r) = \max\left\{0, \min\left\{2r, \frac{1+r}{2}, 2\right\}\right\}. \tag{31}$$

As a generalization, he also suggests to replace the restricted TVD-condition (18) by the general TVD-condition (17). Roe and Baines [5] and Roe [7] point out that it is advantageous to limit the third-order scheme instead of Fromm's scheme in this case. Since there are many limiters

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connected to Roe's name and the resulting limiter is the core of the paper of Arora and Roe [2], we refer to it as the *Arora–Roe-Limiter*:

$$\varphi_{\mathrm{AR}}(r) = \max\{0, \min\{\varphi_{\mathrm{ub}}(r), \varphi_3(r)\}\}$$
$$= \max\left\{0, \min\left\{\frac{2}{|v|}r, 1+\frac{1+|v|}{3}(r-1), \frac{2}{1-|v|}\right\}\right\}. \tag{32}$$

A variant of this, which goes back to van Leer, is to multiply the term $\min\{2, 2r\}$ for φ_{MC} with some $\theta \in [\frac{1}{2}, 1]$. For φ_{AR}, we would in the same way multiply $\varphi_{\mathrm{ub}}(r)$ by θ. Especially in the case of strongly compressive waves, it might be dangerous to use the full freedom given by the original limiter. The bound $\theta \geqslant \frac{1}{2}$ refers to the fact that for equality the MC-limiter just turns into the well-known Minmod-limiter:

$$\varphi_{\mathrm{MC}_{1/2}}(r) = \mathrm{minmod}(r, 1) = \max\{0, \min\{1, r\}\}, \tag{33}$$

which is the lower bound for second-order limiters that vanish for negative r and satisfy condition (M2).

In the following, we employ this idea to construct a variant of the Arora–Roe limiter that allows for the reconstruction of extrema.

A variant allowing for negative values. We now introduce a variant of the Arora–Roe limiter that allows for negative values and, thus, the reconstruction of extrema. By introducing a parameter $\frac{1}{2} \leqslant \theta \leqslant 1$ in the above-mentioned way, one gets some freedom in the TVD-condition. This freedom might be used to extend the area in which third order is obtained to a part of the negative r-axis. For positive r, the limiter satisfies

$$\varphi(r) \leqslant \theta\frac{2}{1-|v|}, \quad \frac{\varphi(r)}{r} \leqslant \theta\frac{2}{|v|}. \tag{34}$$

If for some R equality holds in the first relation of (34), then the general TVD-condition turns into

$$-\frac{2}{1-|v|} \leqslant \frac{\varphi(r)}{r} - \theta\frac{2}{1-|v|} \leqslant \frac{2}{|v|}, \tag{35}$$

from which we conclude

$$-(1-\theta)\frac{2}{1-|v|} \leqslant \frac{\varphi(r)}{r} \leqslant \frac{2}{|v|} + \theta\frac{2}{1-|v|}. \tag{36}$$

Thus, for negative r the limiter has to satisfy

$$\varphi(r) \leqslant -(1-\theta)\frac{2}{1-|v|}r. \tag{37}$$

If for some r equality holds in the second relation of (34), then the general TVD-condition turns into

$$-\frac{2}{1-|v|} \leqslant \theta\frac{2}{|v|} - \varphi(R) \leqslant \frac{2}{|v|}, \tag{38}$$

which leads to the condition

$$\varphi(R) \geqslant -(1-\theta)\frac{2}{|v|}. \tag{39}$$

The optimized limiter, therefore, reads as

$$\varphi_\theta(r) = \min\left\{\max\left\{-(1-\theta)\frac{2}{|v|}, \varphi_3(r)\right\}, \max\left\{-(1-\theta)\frac{2}{1-|v|}r, \theta\frac{2}{|v|}r\right\}, \theta\frac{2}{1-|v|}\right\}. \tag{40}$$

The normal forms of the limiters are presented in Figure 4; numerical results for the application of the limiters of this family are displayed in Figure 5. It can be seen that the Arora–Roe limiter

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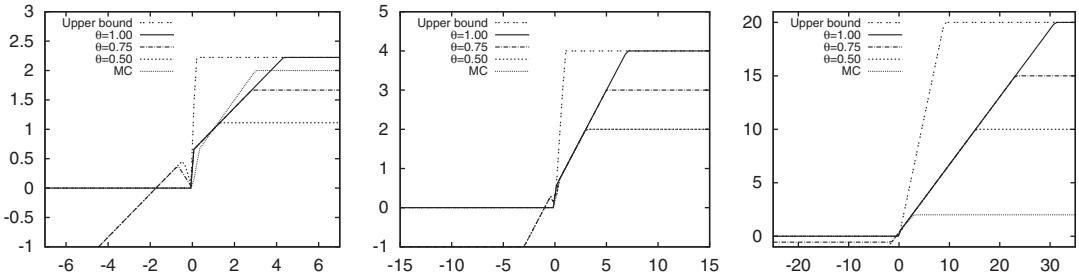


Figure 4. Normal forms of MC, the Arora–Roe limiter ($\theta=1$, and the $\theta=0.75$- and $\theta=0.5$-versions for CFL-numbers 0.1, 0.5, and 0.9.

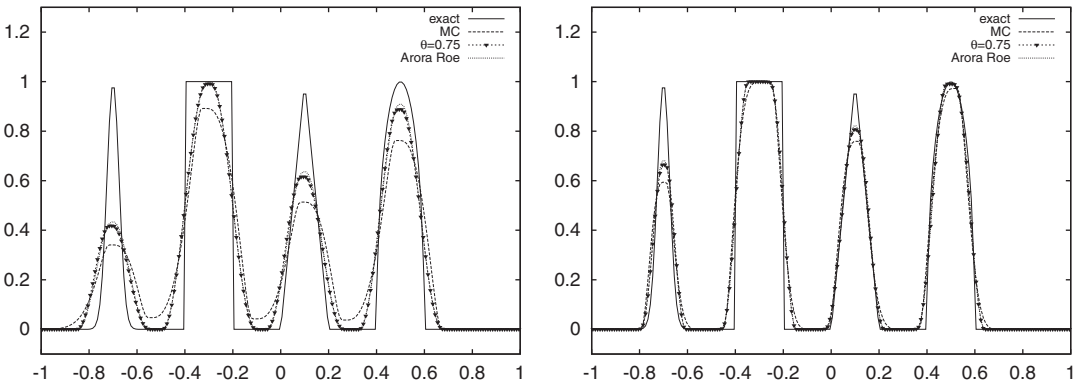


Figure 5. The members of the MC-family applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1 and 0.9.

is superior to the original MC-limiter. The gain of quality by allowing for the reconstruction at extrema, i.e. by using the θ-version with $\theta<1$ instead of the original Arora–Roe limiter, is small compared with the loss due to the restriction in monotone regions. This is mainly due to the fact that in a well-resolved solution only a few extrema occur compared with the full number of grid cells.

3.1.2. The Superbee-family. The starting point for the Superbee-family is the upper-bound limiter φ_{UB} as defined in (22) or, in order to obtain CFL-independent limiters, $\max\{0,\min\{2r,2\}\}$. If in addition to the TVD-condition and the vanishing of φ for negative ratios r inclusion (M2) is enforced, we get in the CFL-dependent case

$$\varphi_{\mathrm{CS}}(r)=\min\{\varphi_{\mathrm{UB}},\max\{1,r\}\}. \tag{41}$$

Roe called this limiter Ultrabee [6]. To avoid confusion with the terminology in [8], we avoid this term[‖] and refer to the limiter as *CFL-Superbee*. In the CFL-independent case, the limiter turns into the well-known Superbee-limiter

$$\varphi_{\mathrm{S}}(r)=\max\{0,\min\{1,2r\},\min\{r,2\}\}. \tag{42}$$

Superbee satisfies (M1)–(M5). The CFL-dependent version just lacks symmetry (M4).

A modification of the CFL-Superbee-limiter can be obtained by the following considerations: For the CFL-number $\frac{1}{2}$, the angle bisector of the graph of φ_{CS} in $r=1$ is perpendicular to φ_3. If this is kept for all CFL-numbers, it may return some of the desirable properties of the third-order

[‖]The confusion with this term culminates in [23] where the authors start with the upper-bound limiter, call it Ultrabee, and modify it to a 'new' limiter called N-Bee, which is just Roe's Ultrabee.

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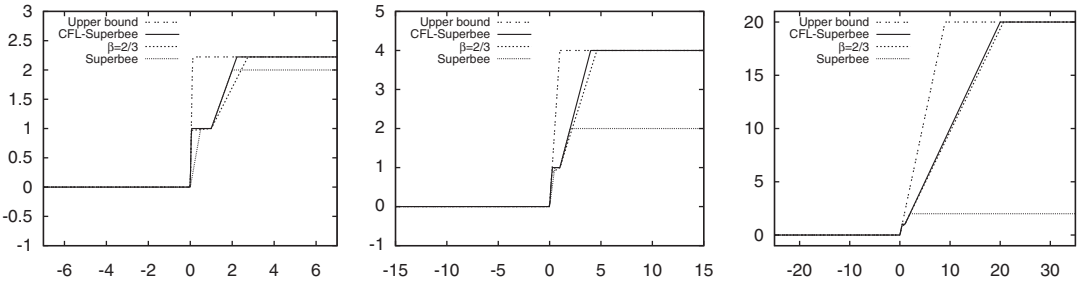


Figure 6. Normal forms of the limiters in the Superbee-family for CFL-numbers 0.1, 0.5, and 0.9.

scheme. With φ'_3 as a shortcut for the slope of the third-order limiter, CFL-Superbee could be modified to

$$\varphi(r)=\max\{0,\min\{\varphi_{\mathrm{ub}}(r),\max\{1+(\varphi'_3-\beta/2)(r-1),1+(\varphi'_3+\beta/2)(r-1)\}\}\}. \tag{43}$$

For the CFL-number $\frac{1}{2}$, the limiter obtained by setting $\beta=1$ would coincide with the original CFL-Superbee. The numerical results are almost the same for both limiters. The differences in the numerical results are small. A difficulty introduced by this modification is the loss of inclusion (M2) and monotonicity (M5) for some β at certain CFL-numbers. While ($\beta>1$) might lead to a strong squaring effect in the numerical results, $\beta\leqslant\frac{2}{3}$ preserves both, inclusion (M2) and monotonicity (M5). Vanishing β would result in the Arora–Roe limiter. The normal forms of the limiters in this section are displayed in Figure 6. The results in Figure 7 show the good approximation of the amplitude by the limiters of this family, especially by the CFL-dependent versions. The choice $\beta=\frac{2}{3}$ results in a slight loss in the maximum at the thin peak and the triangle wave and a slight improvement of the representation of the shape of the half ellipse. The representation of the maxima is still considerably better than that of the CFL-independent standard form of the Superbee-limiter.

3.1.3. The Harten shock switch. The shock switch was introduced by Harten [19] in order to switch to a more compressive limiter at contact discontinuities and steep shear waves. This technique is usually referred to as artificial compression. For this purpose, also other switching functions are available. Most of them are based on a larger stencil. Therefore, we stick to the original switch function of Harten. Its stencil is exactly the same as for the limiters under consideration. Jeng and Payne [24] give an example for the use of a switch with a broader stencil. The normal form of Harten's switching function is

$$\eta(r)=\frac{|1-r|}{1+|r|}. \tag{44}$$

The function vanishes for $r=1$. Hence, it is small in smooth regions. It approaches 1 near discontinuities. If we have two different limiters, one of which is more compressive than the other, e.g.

$$|\varphi_1(r)|\leqslant|\varphi_2(r)| \quad \forall r\in\mathbb{R},$$

then we could create a new optimized one—compressive near discontinuities, less compressive in smooth regions—by the following blending:

$$\varphi(r)=(1-\eta(r))\varphi_1(r)+\eta(r)\varphi_2(r).$$

It is also possible to use the switch to determine an appropriate value for the parameter β in the modified CFL-Superbee-limiter. In the numerical results of this paper a version is included that varies the parameter between $\frac{1}{3}$ and 1:

$$\beta(r)=\tfrac{1}{3}+\tfrac{2}{3}\eta(r). \tag{45}$$

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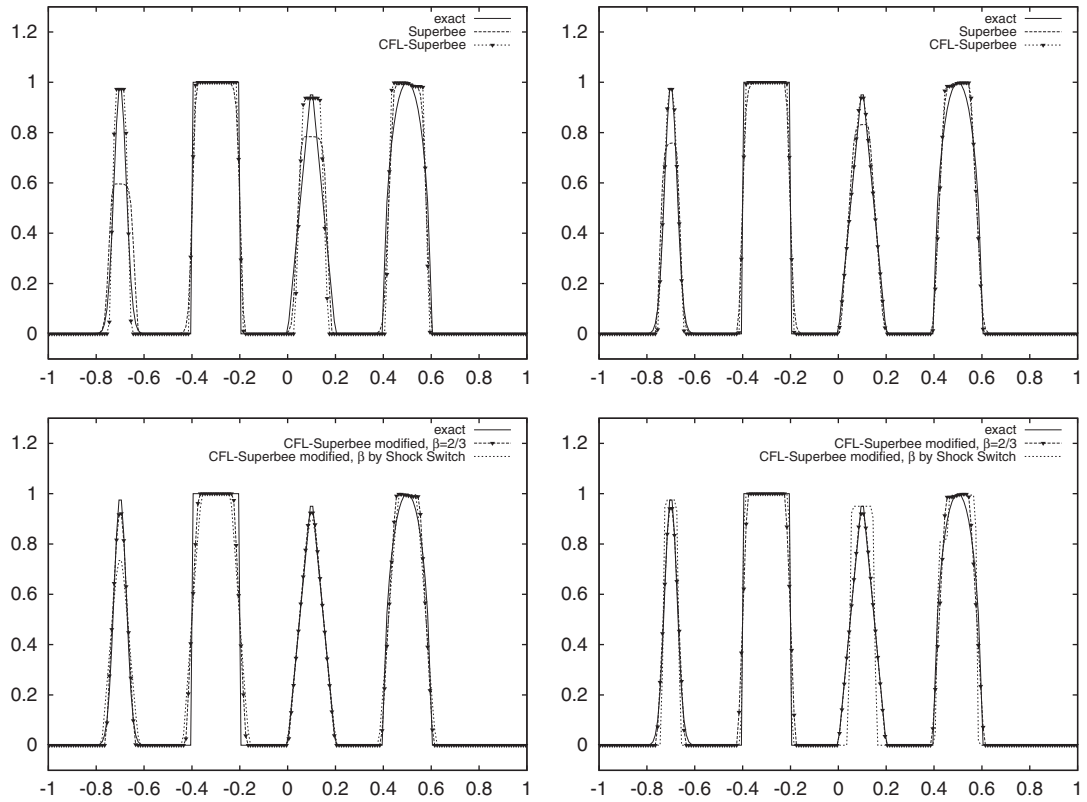


Figure 7. The members of the Superbee-family applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1 and 0.9.

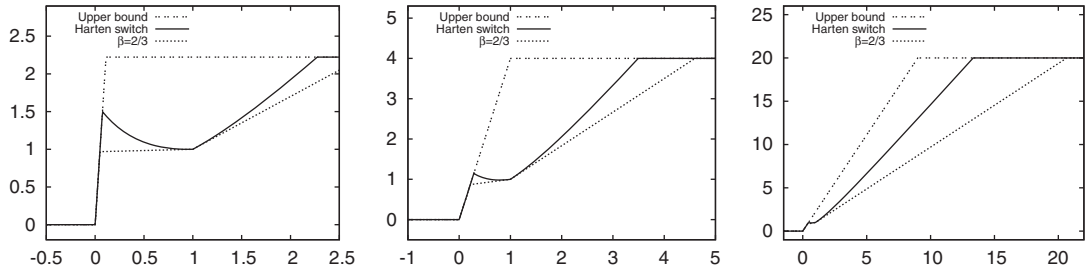


Figure 8. Detail of normal forms of the limiter with the Harten shock switch and with fixed $\beta=\frac{2}{3}$ for CFL-numbers 0.1, 0.5, and 0.9.

This setting turned out to be the best compromise between squaring and viscosity. Although Figure 8 seems to suggest that the limiter with the Harten shock switch is everywhere more compressive than with fixed $\beta=\frac{2}{3}$, it is less compressive in smooth regions. Near $r=1$, condition (45) forces $\beta<\frac{2}{3}$. For the given grid of the Zalesak test in Figure 7, the squaring effect for high CFL-numbers is even stronger than for the original CFL-Superbee. On the other hand, the resolution of the peak and the square wave is poorer than with the fixed $\beta=\frac{2}{3}$.

3.2. Smooth limiters

In this section, we consider limiter functions that are smooth at least on the positive r-axis. Smooth limiter functions are advantageous in simulations that should converge to a steady state [25].

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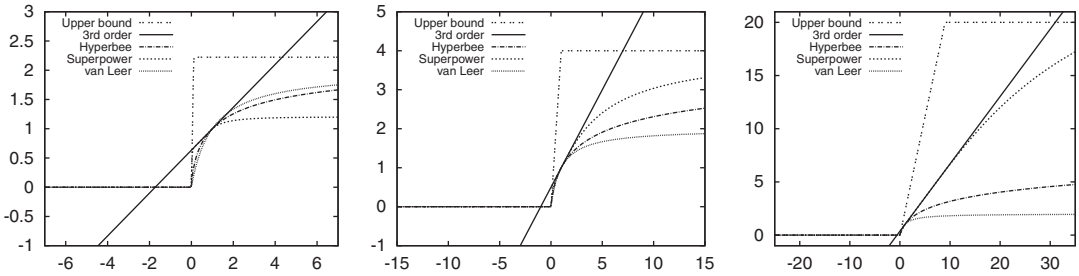


Figure 9. Normal forms of the third-order scheme, Hyperbee, the new Superpower, and the van Leer limiter for CFL-numbers 0.1, 0.5, and 0.9.

Furthermore, they are a good choice for compressive non-linear waves, as can be seen from the results for the Euler equations in Section 5. Prominent examples are the van Leer limiter, which is the application of the harmonic mean and, thus, is smooth on the positive r-axis while vanishing for negative r, and the overall smooth van Albada limiter. For both of these limiters, we show how a complete family of limiters can be made up by using an idea of Serna and Marquina [13]. Another example is the Hyperbee limiter, which was introduced by Roe [7] for theoretical purposes. It was the first CFL-number-dependent third-order adapted smooth limiter. Thus, we put it at the beginning of our considerations.

3.2.1. The hyperbee limiter. The hyperbee limiter was introduced by Roe [7] mainly for theoretical purposes. By using hyperbee, it is possible to advect an exponential initial distribution exactly. The name is due to the fact that it can be written by the use of hypergeometric functions. The normal form reads as

$$\varphi_{\mathrm{hyp}}(r)=\begin{cases}0, & r\leqslant 0;\\ 1, & r=1;\\ \dfrac{2r}{|v|(1-|v|)}\cdot\dfrac{|v|(r-1)+(1-r^{|v|})}{(r-1)^2} & \text{otherwise.}\end{cases} \tag{46}$$

The function is non-decreasing and we find the following limits:

$$\lim_{r\to\infty}\varphi_{\mathrm{hyp}}(r)=\frac{2}{1-|v|},\quad \lim_{r\to 0}\varphi'_{\mathrm{hyp}}(r)=\frac{2}{|v|}. \tag{47}$$

The TVD-condition (17) is sharply satisfied [7]. Furthermore, the slope in $r=1$ coincides with the third-order slope φ'_3. The normal form is displayed in Figure 9 together with the third-order scheme and the members of the van Leer family (cf. next section). The figure shows that for all positive r except $r=1$, they are below the third-order scheme. This makes them good candidates for non-linear waves because they lower the danger of artificial over-compression.

3.2.2. The van Leer family. The name of the van Leer limiter is due to the fact that van Leer already in [26] suggests to employ the harmonic mean in order to get a monotonicity preserving scheme. The normal form of the limiter reads as

$$\varphi_{\mathrm{vL}}(r)=\frac{r+|r|}{1+|r|} \tag{48}$$

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and, thus, vanishes for negative r, is smooth for positive r, and is still continuous in $r=0$. According to Serna and Marquina [13], the limiter can be rewritten as

$$\varphi_{\mathrm{vL}}(r)=\frac{1}{2}(1+\mathrm{sgn}(r))\frac{1+r}{2}\left(1-\left|\frac{1-|r|}{1+|r|}\right|^2\right) \tag{49}$$

$$=\max\left\{0,\frac{1+r}{2}\left(1-\left|\frac{1-|r|}{1+|r|}\right|^2\right)\right\}. \tag{50}$$

This form can easily be generalized by replacing the power 2 by any $p\geqslant 1$. The resulting limiter still satisfies the consistency (M1) or (M1′). Serna and Marquina restrict p to positive integers. But this is not necessary. The analytical results needed in this section are true for all $p>1$. For $p=1$, it is just the Minmod-limiter. For $p>1$ it is smooth on the positive real axis, and we find the following limits:

$$\varphi_{\mathrm{Power}p}(r)\to p \quad (r\to\infty), \tag{51}$$

$$\varphi'_{\mathrm{Power}p}(r)\to p \quad (r\to 0), \tag{52}$$

$$\varphi'_{\mathrm{Power}p}(r)\to \tfrac{1}{2} \quad (r\to 1), \tag{53}$$

$$\varphi''_{\mathrm{Power}p}(r)\to 0 \quad (r\to 0). \tag{54}$$

The derivative $\varphi'_{\mathrm{Power}p}$ is strictly decreasing on the real axis. Since the limiter itself is strictly increasing for positive r, and

$$\varphi''_{\mathrm{Power}p}(1)=0 \quad \forall p>1, \tag{55}$$

we may choose the best possible p (in the sense of the TVD-condition) on both sides of $r=1$:

$$\varphi(r)=\begin{cases}\varphi_{\mathrm{Power}\frac{2}{|v|}}(r), & r\leqslant 1;\\ \varphi_{\mathrm{Power}\frac{2}{|1-v|}}(r), & r\geqslant 1.\end{cases} \tag{56}$$

The resulting limiter satisfies (M1)–(M5). It is also possible to adapt the limiter to the third-order scheme, i.e. replace in (50) the factor $(1+r)/2$ by $\varphi_3(r)$ and adapt p to still satisfy the TVD-condition. This results in

$$\varphi_{\mathrm{Superpower}}(r)=\max\left\{0,\varphi_3(r)\left(1-\left|\frac{1-|r|}{1+|r|}\right|^{p(r)}\right)\right\} \tag{57}$$

with

$$p(r)=\begin{cases}\dfrac{2}{|v|}\cdot 2(1-\varphi'_3), & r\leqslant 1;\\ \dfrac{2}{|1-v|}\cdot 2\varphi'_3, & r\geqslant 1.\end{cases} \tag{58}$$

By this technique we apparently sacrifice symmetry (M4), but still maintain consistency, inclusion, and monotonicity. Figure 10 nicely shows the significant gain in the quality of the numerical results.

3.2.3. Hyperpower, a blending of Hyperbee and Superpower. As can be seen from Figure 9, depending on r and the CFL-number sometimes Superpower is more compressive and sometimes Hyperbee. Therefore, one could use the maximum of them:

$$\varphi_{\mathrm{Hyperpower}}(r)=\max\{\varphi_{\mathrm{Superpower}}(r),\varphi_{\mathrm{hyp}}(r)\}. \tag{59}$$

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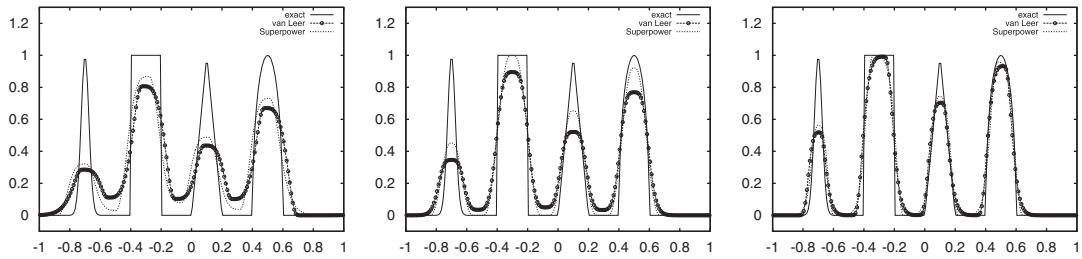


Figure 10. The members of the van Leer-family applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1, 0.5, and 0.9.

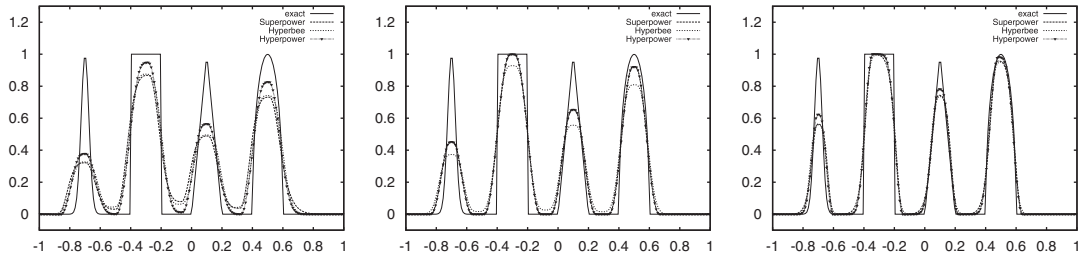


Figure 11. Superpower, Hyperbee, and Hyperpower applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1, 0.5, and 0.9.

According to the results in Figure 11, this leads to a significant enhancement compared with both, Hyperbee and Superpower. While for CFL-number 0.5 it coincides with Superpower, for the other CFL-numbers in consideration the representation of the amplitudes is much better than for the non-blended schemes. Owing to the higher computational cost—both limiters have to be computed and then blended—Hyperpower is of inferior use in practical simulations. It is restricted to situations where a smooth limiter is needed and the cost of the limiter is negligible compared with the cost for the Riemann solver. Therefore, we omit it in our further numerical tests. The search for the optimal smooth limiter is still open.

3.2.4. Klein's sharpening term. Recently, Klein [27] suggested a sharpening term $\psi(\min\{r, 1/r\})$ for the van Leer limiter with

$$\psi(x)=1+x(1-x)(1-x^k), \tag{60}$$

where k is a parameter, usually taken from the positive integers. Hence, the sharpened van Leer limiter reads as

$$\varphi(r)=\varphi_{\mathrm{vL}}(r)*\psi\left(\min\left\{r,\frac{1}{r}\right\}\right). \tag{61}$$

This makes the van Leer limiter into a smooth counterpart of the Superbee-limiter. Now, one could be tempted to apply the same sharpening factor to Superpower. Unfortunately, as Figure 12 shows, the resulting limiter is no longer TVD. Thus, an adapted version of the sharpening is needed. Anyway, we included numerical results for the sharpened Superpower with $k=2$ in the section on the Gresho vortex, Section 5.5.

3.2.5. The van Albada family. The van Albada limiter is motivated by the following considerations. The harmonic mean of a and b is a weighted mean with weights $|b|/(|b|+|a|)$ and $|a|/(|b|+|a|)$. These weights are not differentiable at either $a=0$ or $b=0$, and neither is the resulting mean. To

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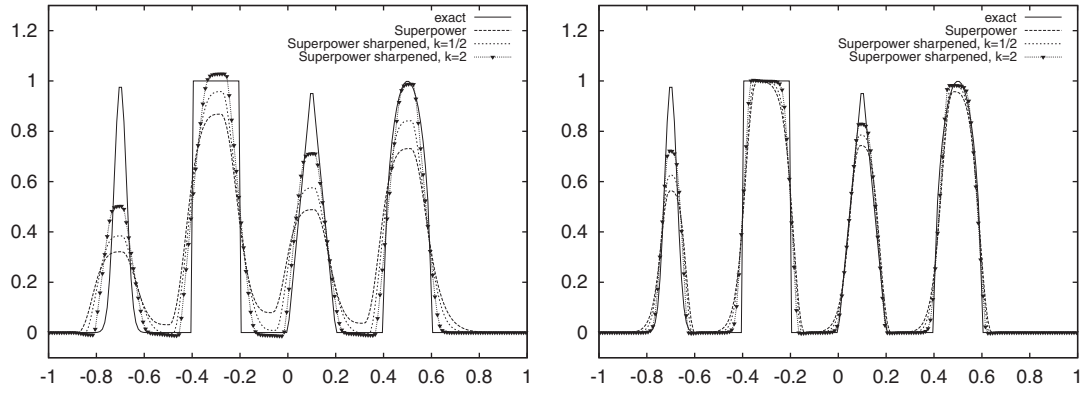


Figure 12. Superpower and the sharpened version for $k=\frac{1}{2}$ and $k=2$ applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1 and 0.9.

overcome this deficiency, van Albada *et al.* [28] suggested the differentiable weights $b^2/(a^2+b^2)$ and $a^2/(a^2+b^2)$. The normal form of the resulting mean is

$$\varphi_{\mathrm{Alb}}(r)=\frac{r+r^2}{r^2+1}, \tag{62}$$

which, obviously, is an overall smooth function. It neither completely vanishes on the negative r axis nor does it use the complete freedom given by the TVD-condition. Actually, it satisfies the sharp condition

$$-\sqrt{2}\leqslant\frac{\varphi_{\mathrm{Alb}}(r)}{r}-\varphi_{\mathrm{Alb}}(R)\leqslant\sqrt{2}. \tag{63}$$

The limiter satisfies (M1)–(M4), but not the monotonicity condition (M5).

Like the van Leer limiter, it can be rewritten as a multiplicative correction of the arithmetic mean:

$$\varphi_{\mathrm{Alb}}(r)=\frac{r+1}{2}\left(1-\frac{(r-1)^2}{r^2+1}\right). \tag{64}$$

If we replace the power 2 by some other power $p\geqslant 1$, we obtain the following generalization:

$$\varphi_{\mathrm{Alb}p}(r)=\frac{r+1}{2}\left(1-\frac{|r-1|^p}{|r|^p+1}\right). \tag{65}$$

Again, for $p=1$ we get the Minmod-limiter. For $p\leqslant 3$, the restricted TVD-condition (18) is satisfied. For $p>1$, the limiter is overall smooth with

$$\lim_{r\to 0}\varphi'_{\mathrm{Alb}p}(r)=\frac{p}{2},\quad \lim_{r\to 1}\varphi'_{\mathrm{Alb}p}(r)=\frac{1}{2}, \tag{66}$$

$$\lim_{r\to 0}\varphi''_{\mathrm{Alb}p}(r)=\frac{3}{2}p-\frac{1}{2}p^2,\quad \lim_{r\to 1}\varphi''_{\mathrm{Alb}\,p}(r)=0. \tag{67}$$

Conditions (M1)–(M4) are still satisfied, monotonicity is only valid for $p=1$ (Minmod). For general $p>1$, the bounds for $\varphi_{\mathrm{Alb}\,p}(r)/r$ and $\varphi_{\mathrm{Alb}p}(R)$ cannot be found directly. An interesting choice is $p=3$. For this limiter, the restricted TVD-condition (18) is sharply satisfied. The derivative is given by

$$\varphi'_{\mathrm{Alb3}}(r)=\frac{1}{2}\left(1-\frac{|r-1|^3}{|r|^3+1}+(r+1)\left[\frac{3r|r||r-1|^3}{(|r|^3+1)^2}-\frac{3(r-1)|r-1|}{(|r|^3+1)}\right]\right). \tag{68}$$

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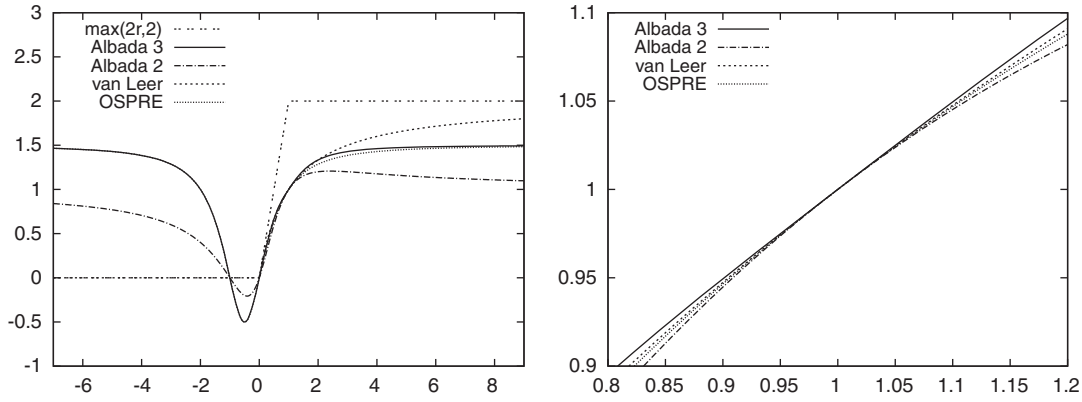


Figure 13. Normal forms (left) and detail (right) of normal forms of Albada 3, Albada 2, van Leer, and OSPRE-limiter.

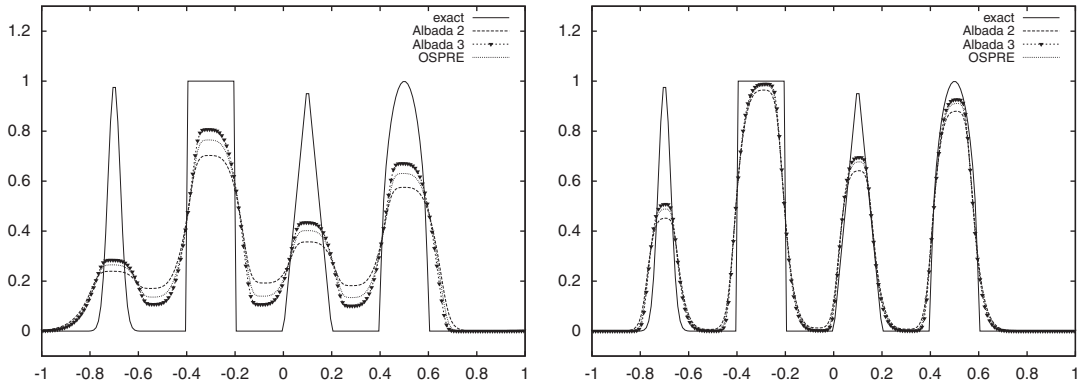


Figure 14. The members of the van Albada-family applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1 and 0.9.

Owing to its independence from the CFL number and smoothness, it might be an interesting alternative to the old limiters in implicit schemes. Since this study is restricted to explicit schemes, we cannot give any information on the performance in the implicit context.

Another approach of generalizing the van Albada Limiter is given by Waterson and Deconinck [12]. Their optimum symmetric polynomial-ratio expression (OSPRE)-limiter

$$\varphi_{\mathrm{OSPRE}}(r)=\frac{3r(r+1)}{2(r^2+r+1)} \tag{69}$$

coincides for negative r with Albada 3, is symmetric, and satisfies the restricted TVD-condition (18) sharply. But, as can be seen from Figure 13, for r close to 1 it is less compressive than Albada 3. Figure 13 shows the normal form of the limiter, together with the original Albada limiter, OSPRE, and the van Leer limiter. The detail presented in the right picture shows that for ratios r close to one, the new Albada 3 limiter is more compressive than the van Leer limiter and, as already mentioned, than OSPRE, which has about the same computational cost. The results in Figure 14 show that the higher computational cost for Albada 3 (in comparison with the original Albada limiter) pays off by a significantly improved resolution, whereas OSPRE does not reach the quality of Albada 3.

Like for the van Leer limiter, it is also possible to create a CFL-number-dependent version. If we define the parameter γ to be the minimum of $2/|v|$ and $2/(1-|v|)$, we find that γ ranges from 2 (for CFL-number zero or one) to 4 (for CFL-number $\frac{1}{2}$). For $p=3$, the term to be bounded for

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the TVD-condition,

$$\tilde{\gamma}=\frac{\varphi_{\mathrm{Alb3}}(r)}{r}-\varphi_{\mathrm{Alb3}}(R),$$

ranges from -2 to 2. By experiments it can be found that for $p=\frac{9}{2}$, the according term ranges from -4 to 4. By setting

$$p(\gamma)=\tfrac{3}{2}+\tfrac{3}{4}\gamma,$$

one could define a limiter

$$\varphi_{\mathrm{Alb3+}}(r)=\varphi_{\mathrm{Alb}\,p(\gamma)}(r).$$

Unfortunately, the relation between $\tilde{\gamma}$ and γ is non-linear. As a consequence, the TVD-condition cannot be guaranteed. The improvement compared with φ_{Alb3} is only visible for CFL-numbers close to $\frac{1}{2}$. Furthermore, the dependency on the CFL-number restricts the use to explicit schemes. All in all, it is only of inferior utility. The most important candidate in the van Albada family is the Albada 3 limiter.

3.3. Double logarithmic limiting and the limiter by Čada and Torrilhon

In [16], Čada and Torrilhon present an example of a modern non-TVD-limiter. The starting point of the construction is the double logarithmic limiter by Artebrant and Schroll [15], the normal form of which, according to [16], can be written as

$$\varphi_{\mathrm{dlog}}(r)=\frac{2p[(p^2-2pr+1)\log p-(1-r)(p^2-1)]}{(p^2-1)(p-1)^2} \tag{70}$$

with

$$p=p(r)=2\frac{|r|^q}{1+|r|^{2q}} \tag{71}$$

and a parameter $q<1$. The occurring singularities are removable with $\varphi_{\mathrm{dlog}}(1)=1$, $\varphi_{\mathrm{dlog}}(-1)=\frac{1}{3}$, and $\varphi_{\mathrm{dlog}}(0)=0$. While violating the TVD-condition in general, it is still total variation bounding in some local sense [15]. The limiter is designed for the use in semi-discrete schemes. Since the reconstruction step in this context corresponds to CFL-number zero—nothing is transported—schemes of form (9) show up third-order behaviour only in semi-discrete schemes or for small CFL-numbers. The quality of the limiter varies with q on different regions of the r-axis. To resemble the best properties for the different values of q and, at the same time, reduce the computational cost, Čada and Torrillhon suggest, for semi-discrete schemes, the following limiter:

$$\varphi_{\check{\mathrm{C}}\mathrm{T}}(r)=\max\left\{\frac{-3}{|r|},\min\left\{\frac{2+r}{3},\frac{2|r|}{0.6+|r|},\frac{5}{|r|}\right\}\right\}. \tag{72}$$

Especially in the vicinity of $r=0$, the TVD-condition is violated. To adapt the limiter to fully discrete schemes and still get third order, the term $(2+r)/3$ should be replaced by $\varphi_3(r)$. By using the results of Section 2.3, it is possible to create a version that satisfies the TVD-condition (17) sharply at least for r, $R\geqslant 0$ or r, $R\leqslant 0$. We can use the freedom in (17) to allow for higher absolute values of $\varphi(r)$. To do this, we tune the parameters in (72) so that the basic structure of the limiter is preserved and (17) is sharply satisfied as long as r and R have the same sign. The resulting limiter, then, reads as

$$\varphi_{\check{\mathrm{C}}\mathrm{Tmod}}(r)=\max\left\{\frac{-2}{|r|}\cdot\frac{|v|^2-3|v|+8}{|v|^3-|v|^2-|v|+1},\min\left\{\varphi_3(r),\frac{2|r|}{|v|},\frac{1}{|r|}\cdot\frac{8-2|v|}{(|v|-1)^2}\right\}\right\}. \tag{73}$$

Figure 15 shows the normal forms of the Čada–Torrilhon limiter and the new modification for different CFL-numbers. The compression allowed by the modification is much higher than for

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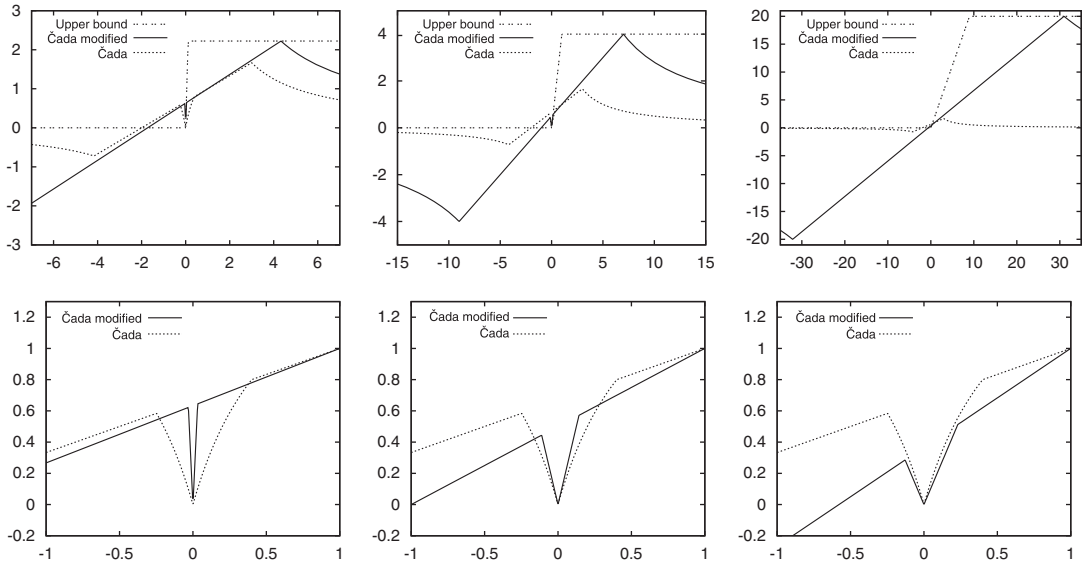


Figure 15. Normal forms (upper row) and detail (lower row) of normal forms of the limiter by Čada and Torrilhon and the new modification for CFL-numbers 0.1, 0.5, and 0.9.

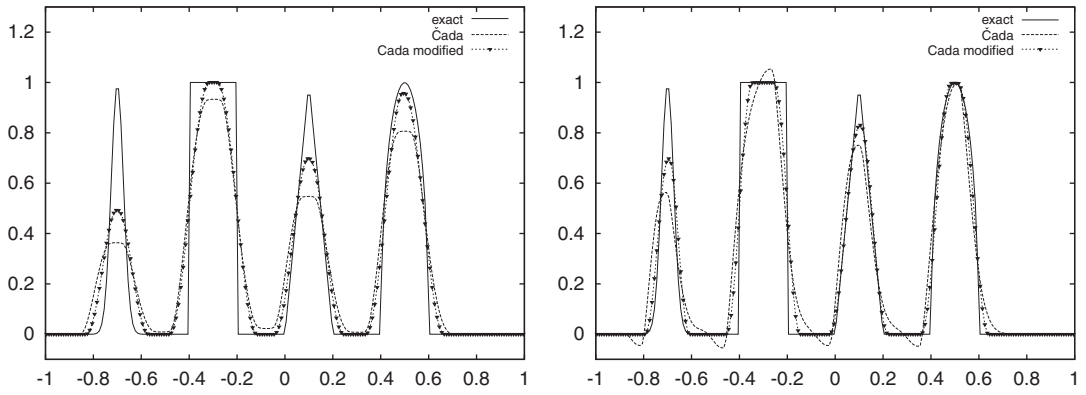


Figure 16. The Čada–Torrilhon limiter and its modification applied to the Zalesak test problem at time $t=20$ for CFL-numbers 0.1 and 0.9.

the original limiter. The detail view in the lower row of Figure 15 shows that for small CFL-numbers, the modification sticks much longer to the third-order scheme. For higher CFL-numbers, the deviation of the original limiter from third order increases considerably.

In practical use, one should multiply all parts besides $\varphi_3(r)$ by $(1-\varepsilon)$ with some small parameter ε, ranging between 10^{-1} and 10^{-3}. To avoid division by zero, the denominators depending on the CFL-number ν should be replaced by $|\nu|^3-|\nu|^2-|\nu|+1+\varepsilon$, $|\nu|+\varepsilon$, and $(|\nu|-1-\varepsilon)^2$. The results displayed in Figure 16 clearly show the improvement gained by this modification. They also show that it is not reasonable to use a limiter constructed for the use in semi-discrete schemes in a scheme of the form (9). Therefore, we omit the original Čada-limiter in our further comparisons.

For compressive non-linear waves, we use an intermediate form to avoid over-compression. For the leftmost and the rightmost parts, we use the original setting of Čada and Torrilhon. The middle part is taken as for $\varphi_{\check{\mathrm{C}}\mathrm{Tmod}}(r)$. The resulting limiter reads as

$$\varphi_{\check{\mathrm{C}}\mathrm{T\ intermediate}}(r)=\max\left\{\frac{-3}{|r|},\min\left\{\varphi_3(r),\frac{2|r|}{|\nu|},\frac{5}{|r|}\right\}\right\}. \tag{74}$$

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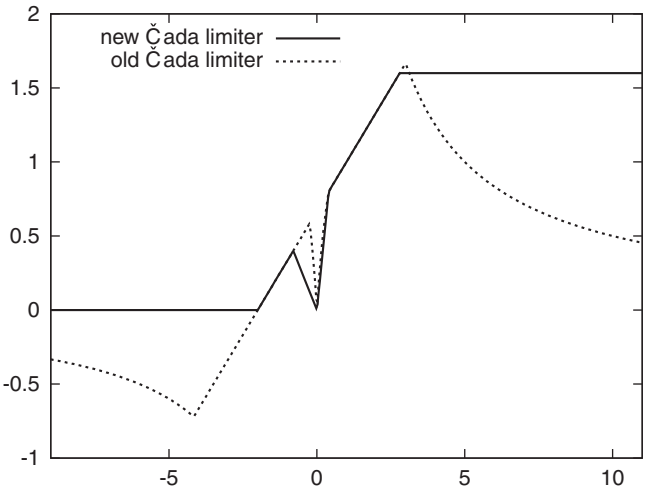


Figure 17. Comparison of new and old Čada limiters.

In their last paper [29], Čada and Torrilhon suggest a new limiter [29], which is a member of the MC-family and, thus, deviates significantly from the double logarithmic limiter. As can be seen from Figure 17, it is a CFL-number-independent variant of the $\theta=0.8$ limiter and, thus, far away from the double logarithmic limiter. Therefore, in the remainder of the paper, we only consider the old one.

3.4. The Čada switch

In their above-mentioned paper [29], Čada and Torrilhon present a new switch function that tests for smooth extrema. This is independent from the limiter in use as long as the limiter is based on a third-order scheme. It was originally used in connection with their old limiter (72), cf. [30]. The goal is to single out the cases in which, although being in the vicinity of $r=0$, we might safely switch to the third-order scheme, i.e. the cases in which $\varphi(0)\neq 0$ causes no troubles. The indicator for smooth extrema is defined as

$$\eta(\Delta q_{\text{up}}, \Delta q_{\text{down}}) = \frac{\Delta q_{\text{up}}^2 + \Delta q_{\text{down}}^2}{(\text{rad}\Delta x)^2} \tag{75}$$

with a suitable chosen 'asymptotic radius' rad. Naively used, this leads to a violation of homogeneity (M3). The scheme would depend on the reference values used to get the non-dimensional form of the physical system. The asymptotic radius should depend on the state, e.g., and some length, e.g. in the case of 1d linear advection some constant times total variation of the initial state over the size of the computational domain. Unfortunately, Čada and Torrilhon give no hint on how to choose this radius. For the purpose of our comparison, we simply choose $\text{rad}=0.1$, which seems to be reasonable for the test cases under consideration. They apply the indicator to construct a smooth switch between the original limiter and the third-order scheme in the following manner:

$$\varphi_{\text{switch}}(r) = \begin{cases} \varphi_3(r), & \eta \leqslant 1-\varepsilon; \\ \varphi_{\text{orig}}(r), & \eta \geqslant 1+\varepsilon; \\ \dfrac{1}{2}\left(\left(1-\dfrac{\eta-1}{\varepsilon}\right)\varphi_3(r) + \left(1+\dfrac{\eta-1}{\varepsilon}\right)\varphi_{\text{orig}}(r)\right) & \text{else} \end{cases} \tag{76}$$

with some small $\varepsilon>0$; in our tests $\varepsilon=0.1$. The resulting scheme is no longer TVD as is confirmed by the numerical examples in Figure 18. The results also show that the scheme may destroy the shape of the waves. Thus, the switch should only be applied with care until general criteria are found for a reasonable choice of the parameters involved.

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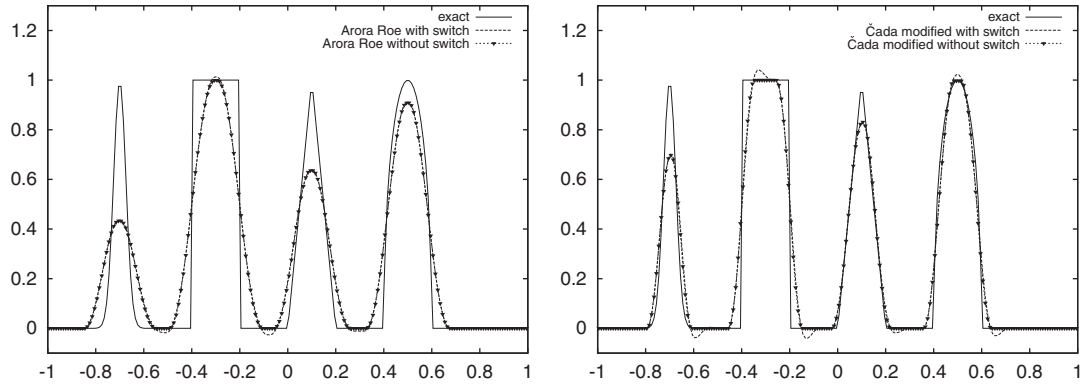


Figure 18. Effect of the Čada switch on Arora–Roe for CFL-number 0.1 (left) and the modified Čada–Torrilhon limiter for CFL-number 0.9 (right).

4. CONVERGENCE STUDY

To test the convergence of the schemes resulting from the different limiters, we employ the function $\sin(6\pi x)$ as initial data on the interval $[-1, 1]$, i.e. a wave packet consisting of six periods. The computations are done with $T = 8$. Since this corresponds to four full rounds, it is possible to use the initial data as the exact solution to which we have to compare the computed one. The error is measured in the normalized $\mathcal{L}_2$-norm. The grid resolution starts with 100 grid cells. By doubling the number of grid cells, we go up to 25 600 grid cells. All computations are done for the CFL-numbers 0.1 and 0.9. First we test the original schemes, i.e. the application of the limiters without the Čada switch. Then, we show how the switch affects the convergence of the schemes.

4.1. Results without the Čada switch

First we test the limiters in their pure form, i.e. without any additional switch. Figure 19 shows the results for the CFL-independent limiters. The most striking result is the poor performance of the Superbee-limiter. Owing to the squaring effect, it is only slightly better than Minmod on fine grids. On coarse grids, it still gives good results. Owing to the clipping of minima and maxima, none of the schemes resulting from CFL-independent limiters shows full second order. Even the Albada 2 and Albada 3 schemes, although allowing for reconstruction in some extrema, do not show second order. Near $r = 0$, they also clip local minima and maxima.

According to Figure 20, the same is true for the limiters that make use of the local CFL-number. The only scheme showing full third-order convergence is the linear third-order scheme. The Arora–Roe scheme, which shows the best convergence of all limited schemes, is close to third order on coarse grids. On fine grids, it is slightly below second order. The effects of the clipping overweigh the third-order behaviour near $r = 1$. The $\beta = 2/3$-limiter, although adapted to the third-order slope at $r = 1$, still suffers from the squaring effect. The results for Hyperpower, which is the best smooth candidate, are almost the same as for the $\theta = 0.5$-limiter, which is the poorest CFL-dependent piecewise linear candidate.

4.2. Results with Čada switch

Now we investigate the influence of the Čada switch on the convergence of the schemes. For most limiters, no difference or only a small difference can be seen. The closer the underlying scheme to third order, the earlier the effect of the switch can be seen. Figure 21 shows that also some unexpected effects may occur. For the Arora–Roe scheme, the error, probably due to rounding errors, first jumps up before it drops down to the error of the linear third-order scheme. For the modified Čada limiter, we can see that the error at some resolution pretty fast changes from the original one to the linear third-order scheme. Since the effect is only seen on really fine grids and

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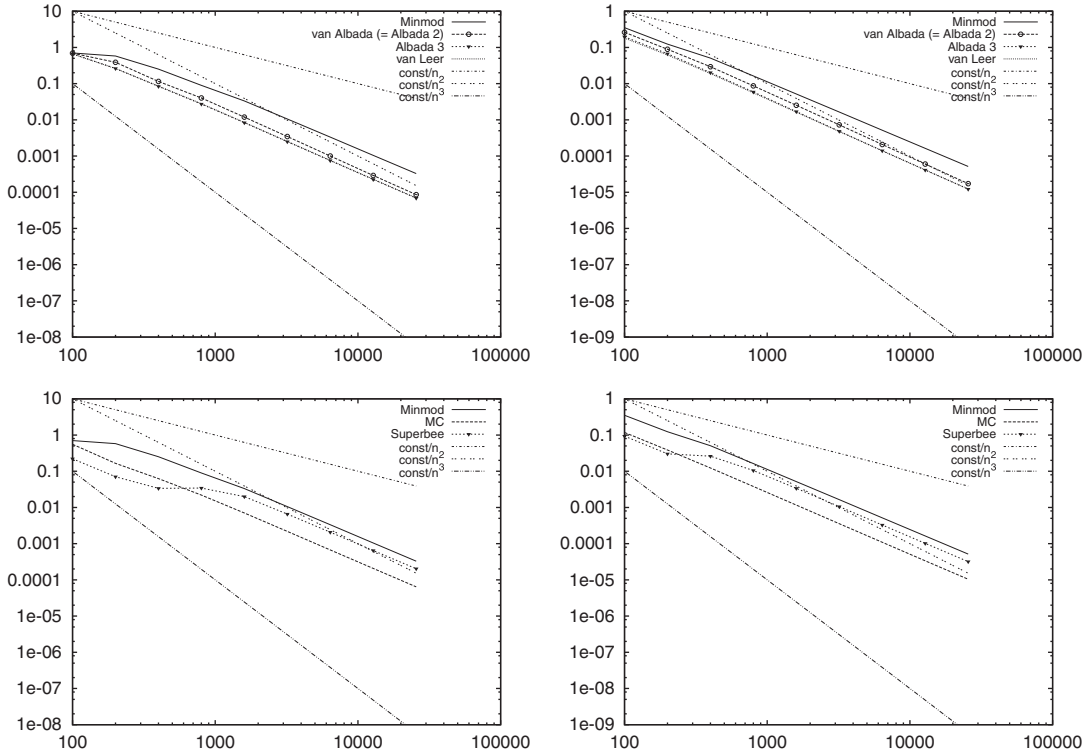


Figure 19. Convergence of CFL-number-independent limiters for sinusoidal wave; upper row smooth, lower row piecewise linear; Minmod as point of comparison; left to right: CFL-numbers 0.1, 0.5, and 0.9.

since the problem of the choice of the parameters is not yet solved, the switch should be only applied with extreme caution.

4.3. Consequences for the use of limiters

At this point, we may draw the first consequences from the numerical results although we have only considered the case of linear advection.

On coarse grids, compressive limiters such as Superbee, CFL-Superbee, and the $\beta=\frac{2}{3}$ limiter yield the best results. Especially piecewise linear data are transported almost perfectly over long times by using CFL-Superbee. The $\beta=\frac{2}{3}$ limiter turns out to be the best compromise between the resolution of piecewise linear and smooth data. For smooth data on fine grids, the Arora–Roe solver is the best choice. It does not suffer from squaring effects like the members of the Superbee-family and, away from local maxima and minima, resembles the linear third-order scheme.

If the limiter is required to be overall smooth, Albada 3 is the best choice. If a kink in $r=0$ is allowed, Hyperbee and Superpower are fine candidates. Hyperpower, which is kind of a blending of both, is almost smooth** and yields even better convergence results.

The results of this section are obtained for the linear advection equation. In the application to non-linear systems like the Euler equations, we expect the same properties to be found for linearly degenerated waves like entropy- and shear-waves. The question on the behaviour for genuinely non-linear waves cannot be answered by these results. Therefore, in the next section we test the limiters with different standard settings for the Euler equations.

**In Section 3.2.3, we defined it to be the maximum of Superpower and Hyperbee. Thus, it is not smooth for $r\geqslant 0$, but only has very weak kinks.

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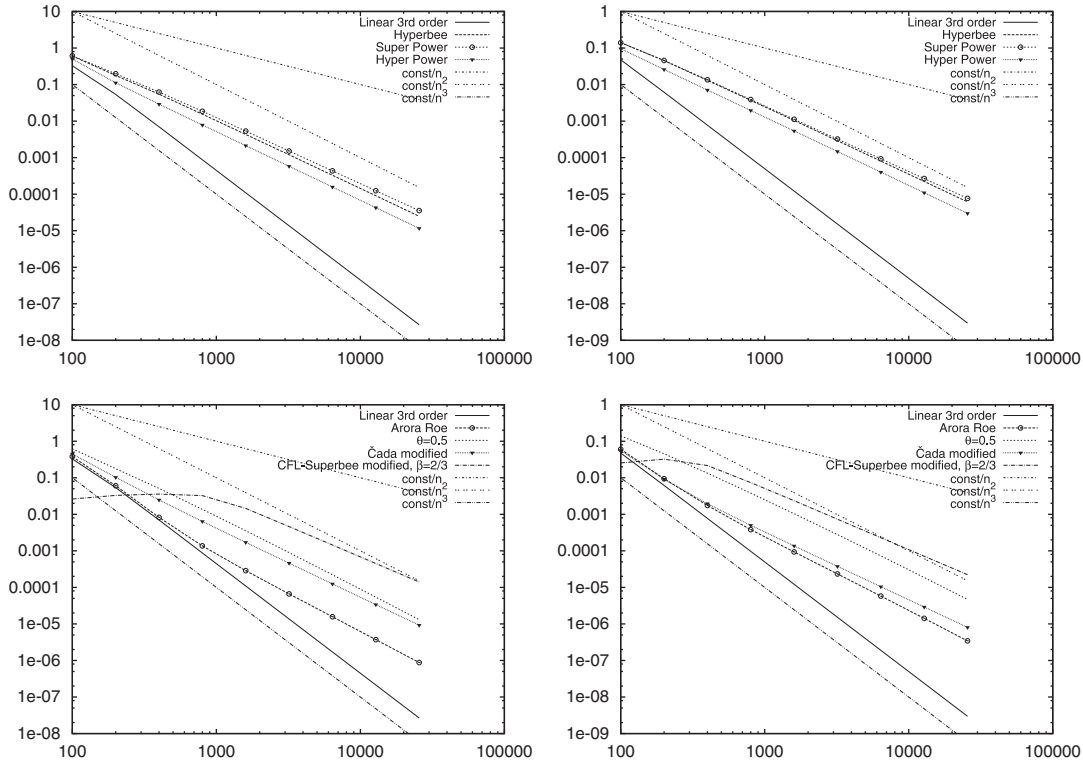


Figure 20. Convergence of CFL-number-dependent limiters for sinusoidal wave; upper row smooth, lower row piecewise linear; Minmod as point of comparison; left to right: CFL-numbers 0.1 and 0.9.

5. NUMERICAL RESULTS FOR THE EULER EQUATIONS

This section presents numerical results for the Euler equations. All plots are done for the density since it is the only variable that shows all waves, at least in one space-dimension. We show results for the following test problems: the blast wave problem of Woodward and Colella [31], the shock–entropy-wave interaction problem of Shu and Osher [32], a 2d Riemann problem, the implosion problem given in [8], and the Gresho vortex.

5.1. The blast wave problem of Woodward and Colella

The blast wave problem of Woodward and Colella is chosen because it includes very compressive non-linear waves, and, thus is very sensitive to compressive limiters. The problem consists of two colliding blast waves of different strength. It is set in the interval $[0, 1]$ with the blast waves starting at $x = 0.1$ and $x = 0.9$. In the initial state, the density is constant set to one, the velocity is zero, and the pressure in the three areas is

$$p_l = 10^3, \quad p_m = 10^{-2}, \quad p_r = 10^2. \tag{77}$$

The reference solution is done with the $\theta = 0.75$ version of the Arora–Roe-limiter on a 25 600 grid.

Owing to the highly compressive waves, one might expect the compressive limiters like CFL-Superbee or Superbee to cause problems since the resulting scheme could then be, in some sense, overly compressive. The results in Figure 22 show that even the Superbee, the most compressive CFL-independent limiter does not suffer from its compressive character. The comparison of the CFL-dependent schemes in Figures 23 and 24 shows that not only CFL-Superbee but almost all limiters that use the full freedom of the general TVD-condition (17) show up some kinds of

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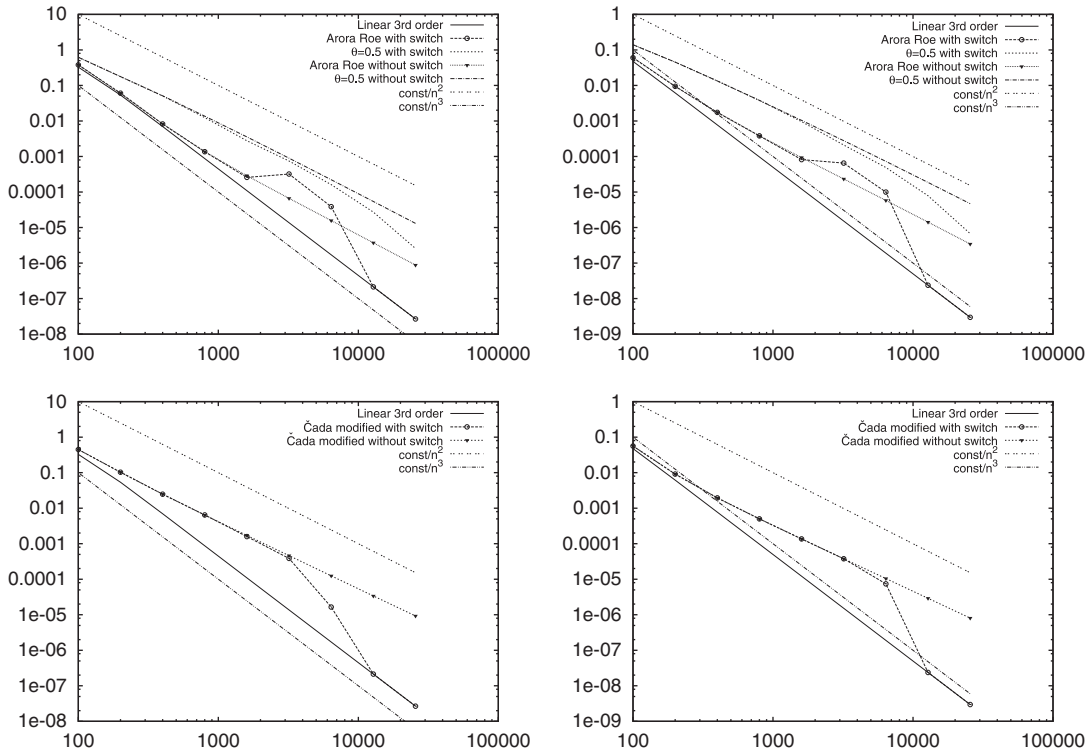


Figure 21. Effect of the Čada switch on the convergence; Arora–Roe and $\theta=0.5$ (upper row) and the modified Čada limiter (lower row) for CFL-numbers 0.1 (left) and 0.9 (right).

overshoots and oscillations. This includes CFL-Superbee, its modified version with $\beta=\frac{2}{3}$, Arora–Roe, and the modified version of the limiter by Čada and Torrilhon. The smooth limiters Hyperbee and Superpower do not suffer. The problems for the Arora–Roe scheme are much smaller than for the CFL-Superbee. For the θ-version of Arora–Roe, the results are free of these deficiencies. Unfortunately, this is at the cost of much higher numerical viscosity. Hence, the problem is not so much the behaviour near $r=1$ as the amount to which the freedom of the TVD-condition (17) is used.

Should one, therefore, resort to a CFL-independent? Is it necessary to abandon the full freedom of the general TVD-condition (17)? The answers are given in Figure 25. For the linear wave, we employ CFL-Superbee. This is a reasonable choice because the solution is piecewise linear. Thus, we expect no squaring effect. For the non-linear waves, we tested Superbee, the $\theta=0.75$ version of Arora–Roe, and the modified Superpower. Superbee, which is not adapted to the third-order scheme in $r=1$, gives a poorer resolution than the other schemes. On the coarse grid, it can be seen that the shape of the right peak is weakly reproduced. The other two schemes give identical results, although Superpower uses the full freedom of the general TVD-condition (17).

The results in Figure 25 are much better than the results obtained by Serna and Marquina [13] with their fifth-order WPower-ENO5, underlining the gain in quality by taking into account the local CFL-number.

5.2. The shock–entropy-wave interaction problem of Shu and Osher

The shock–entropy-wave interaction problem of Shu and Osher [32] involves a moving shock interacting with density fluctuations of small amplitude. The initial data are

$$\rho=3.857143, \quad u=2.629369, \quad p=10.333333 \quad \text{for } x\in[-5,-4), \tag{78}$$

$$\rho=1+0.2\sin 5x, \quad u=0, \quad p=1 \quad \text{for } x\in[-4,5]. \tag{79}$$

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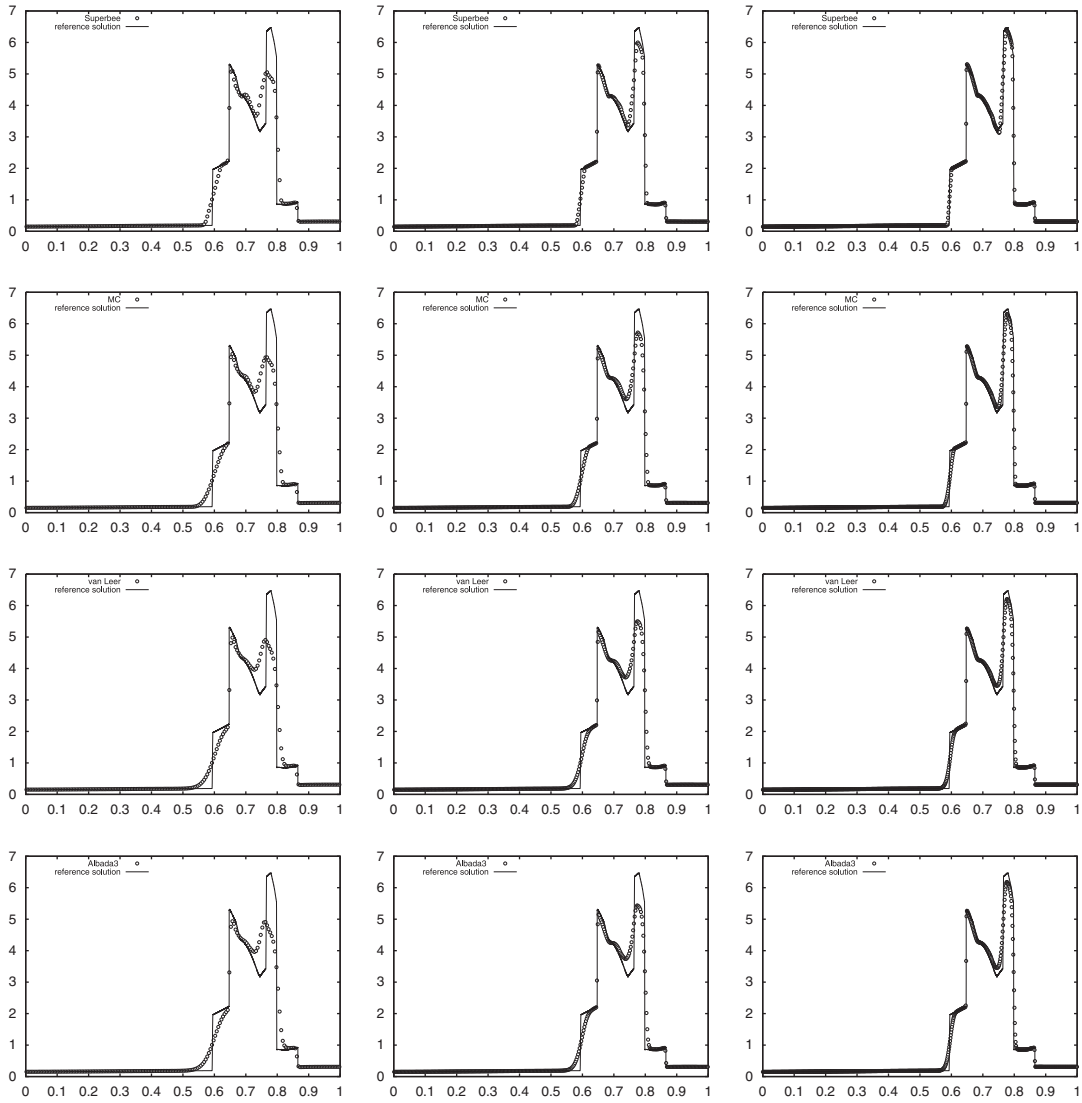


Figure 22. Woodward–Colella blast wave problem at $t=0.038$; results CFL-independent limiters: Superbee, MC, van Leer, Albada 3 (top down); resolution: 200, 400, and 800 cells (left to right).

Like in [32], results are presented for $t=1.8$. Since the waves in this problem are much less compressive than for the blast wave problem, it is possible to apply all limiters to all waves. Where not stated otherwise, in our computations we used the same limiter for the non-linear waves as for the linear entropy-wave. The computational domain was resolved by 400 grid cells on the interval $[-5, 5]$. The depicted results show a zoom of the interaction region. Figure 26 shows the results for the CFL-independent schemes. Although Superbee gives a better approximation of the amplitude than the others, its resolution is not yet satisfying. The results for the third-order adapted schemes in Figure 27 show that Hyperbee introduces even more numerical viscosity than the θ-limiter with $\theta=0.5$. Superpower provides an alternative in the context of smooth limiters, which gives a considerable improvement. But still the numerical viscosity is slightly higher than for the θ-limiter with $\theta=0.75$. The best results, as can be seen in Figure 28, are obtained with Roe's CFL-Superbee and its variations. The use of the modification with $\beta=\frac{2}{3}$ only leads to a minimal loss in amplitude. Surprisingly, the best results are obtained when the Superpower limiter is applied to the non-linear waves. This might be a consequence of the reduced squaring effect.

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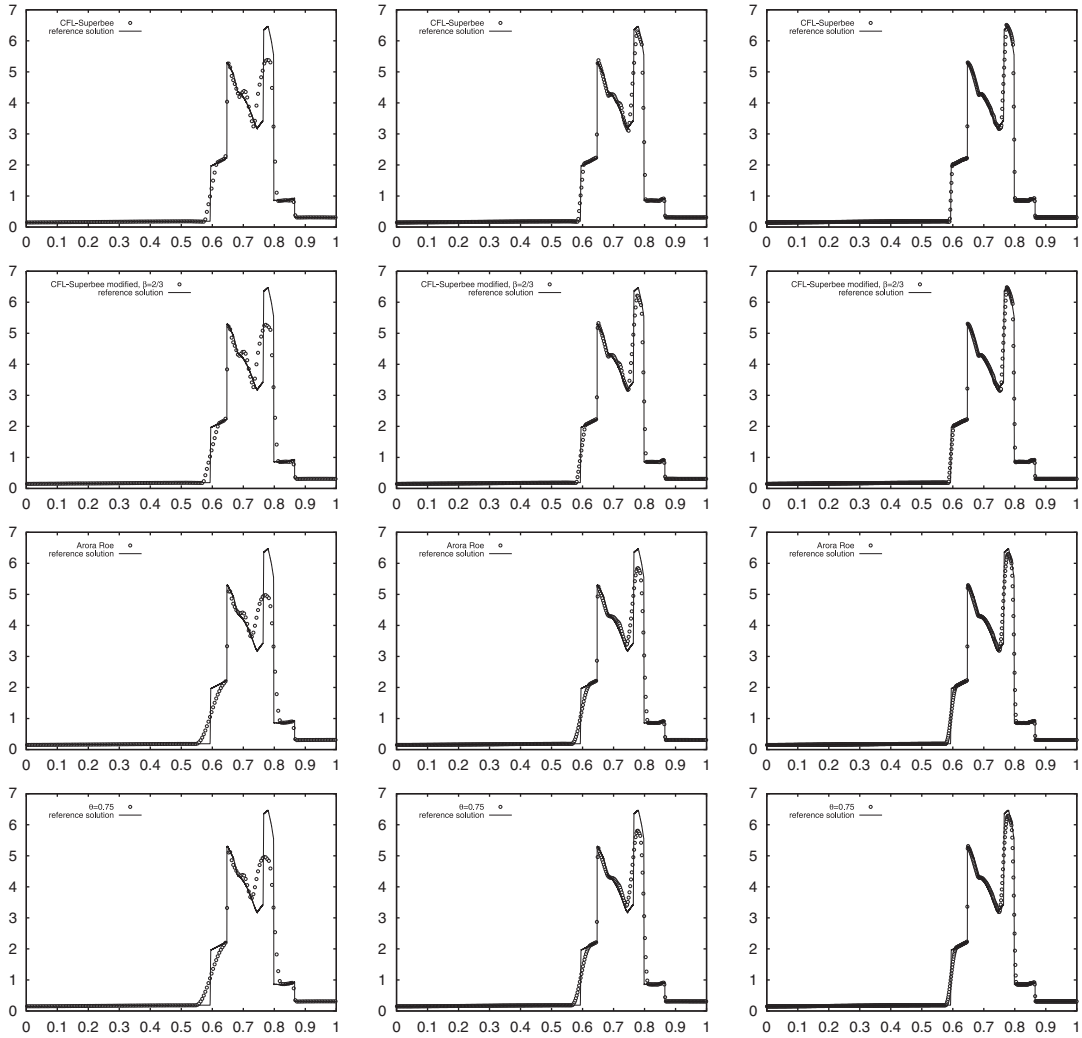


Figure 23. Woodward–Colella blast wave problem at $t=0.038$; results CFL-dependent piecewise linear limiters: CFL-Superbee, modified version with $\beta=\frac{2}{3}$, Arora–Roe, version with $\theta=0.75$ (top down); resolution: 200, 400, and 800 cells (left to right).

Again, the results are better than those obtained by Serna and Marquina [13] with WPower-ENO5.

5.3. Two-dimensional Riemann problem

As an example of a 2d simulation, we show a 2d Riemann problem. This is one of the examples introduced in [33] and frequently used as a test for numerical schemes, e.g. [10, 34, 35]. The initial states in the four quadrants, ordered counterclockwise starting with the upper right quadrant, are

$$\rho=1.5, \quad u=0, \quad v=0, \quad p=1.5,$$

$$\rho=0.532258064516129, \quad u=1.206045378311055, \quad v=0, \quad p=0.3,$$

$$\rho=0.137992831541219, \quad u=1.206045378311055, \quad v=1.206045378311055,$$

$$p=0.029032258064516,$$

$$\rho=0.532258064516129, \quad u=0, \quad v=1.206045378311055, \quad p=0.3.$$

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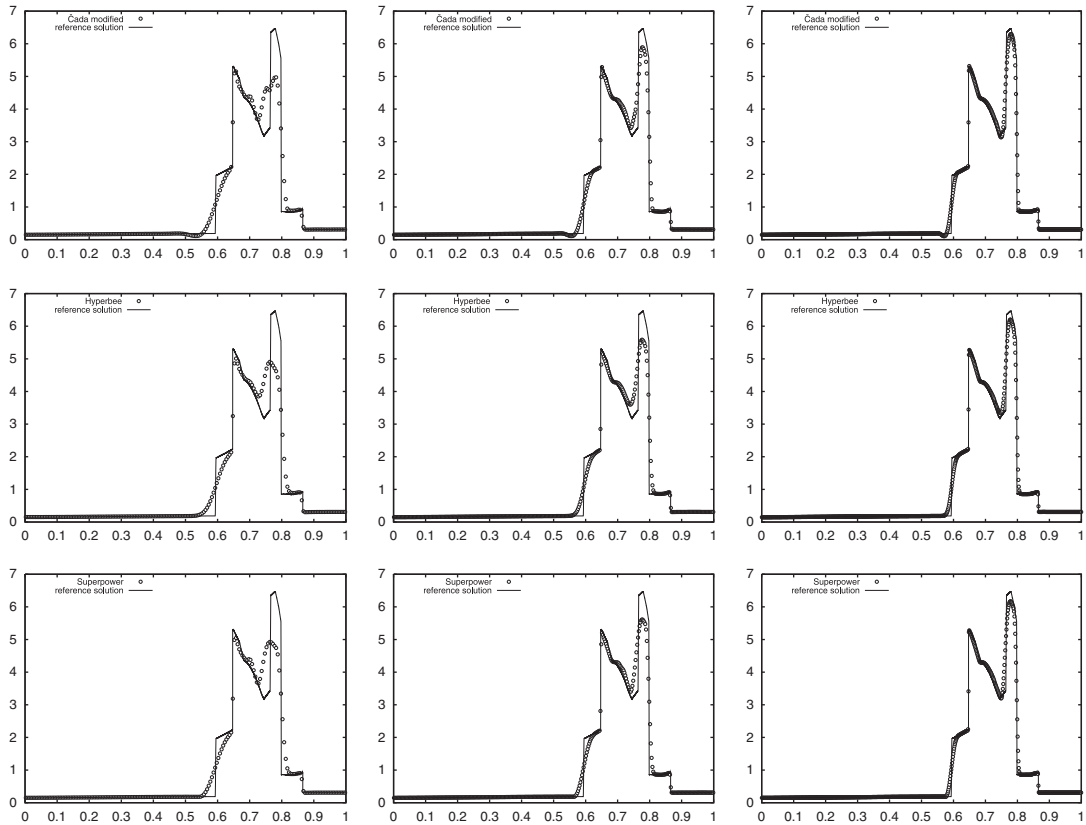


Figure 24. Woodward–Colella blast wave problem at $t=0.038$; results CFL-dependent limiters, which are not piecewise linear: Čada–Torrilhon modified, Hyperbee, Superpower (top down); resolution: 200, 400, and 800 cells (left to right).

In Figure 29, we show results on a 200×200 grid. For the CFL-independent cases, we used the same limiter on all waves. For the other computations, we used Superpower on the non-linear waves and a more compressive one on the linear waves as indicated in the caption of Figure 29. The reference solution is done with the Arora–Roe limiter on a 800×800 grid. The results show that it is possible to find a great deal of the structure of complex problems already on a relatively coarse grid just by switching to a CFL-number-dependent limiter. The CFL-dependent limiters reveal more structure than the schemes tested in the above references although most of them used finer grids.

A drawback of this example is the lack of information on the exact solution. It is inherently unstable. Therefore, we add two examples for 2d Euler flow whose solution involves some radial symmetry. For one of them, the implosion problem, we can give a reference solution that is done with a 1d-code. For the other one, the exact solution is known.

5.4. An implosion problem

In his textbook [8], Toro gives some radial symmetric test problems. One of them is an implosion problem with initial velocity set to zero. The density and pressure inside a circle of radius 0.4 are set to

$$\rho_{\text{in}}=0.125,\quad p_{\text{in}}=0.1.$$

Outside of the circle, density and pressure are both set to 1. The radial symmetry allows for a reference solution done with a 1d-code and geometric source terms. For the tests presented in Figure 30, we applied the same limiter to all waves. It can again be seen that a CFL-dependent

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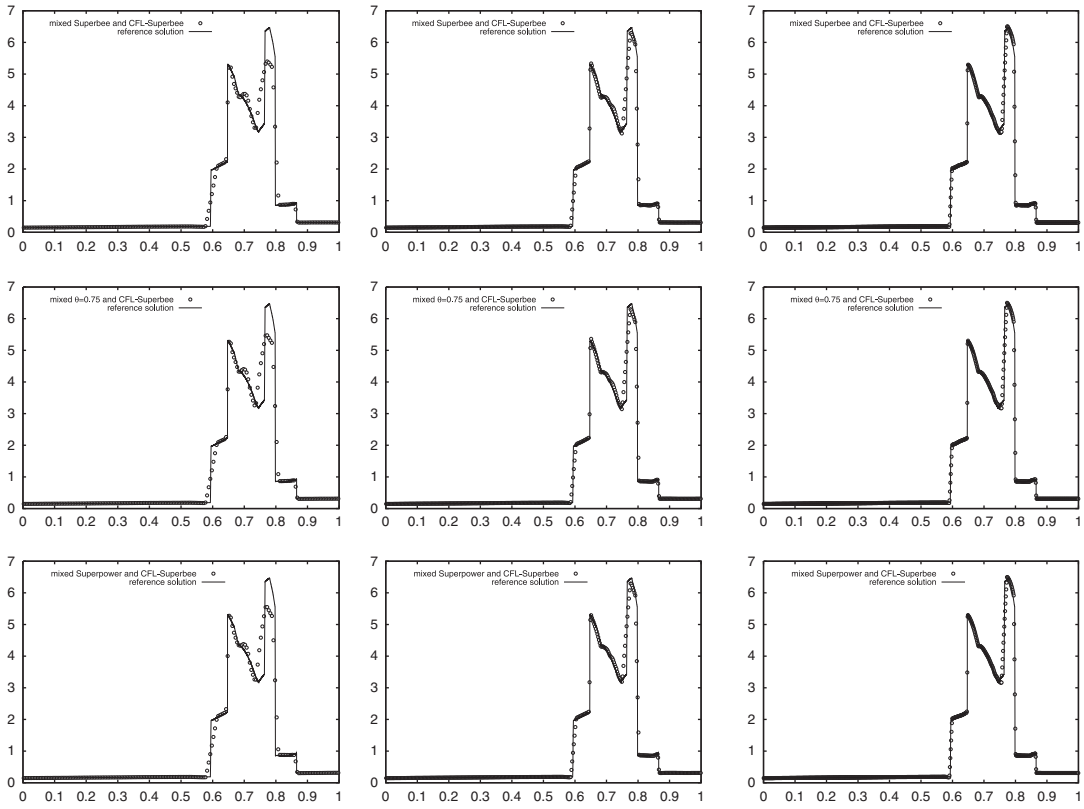


Figure 25. Woodward–Colella blast wave problem at $t = 0.038$; results for mixed use of limiters: CFL-Superbee for contact discontinuity, for non-linear waves: Superbee, θ-limiter (40) with $\theta = 0.75$, Superpower (top down); resolution: 200, 400, and 800 cells (left to right).

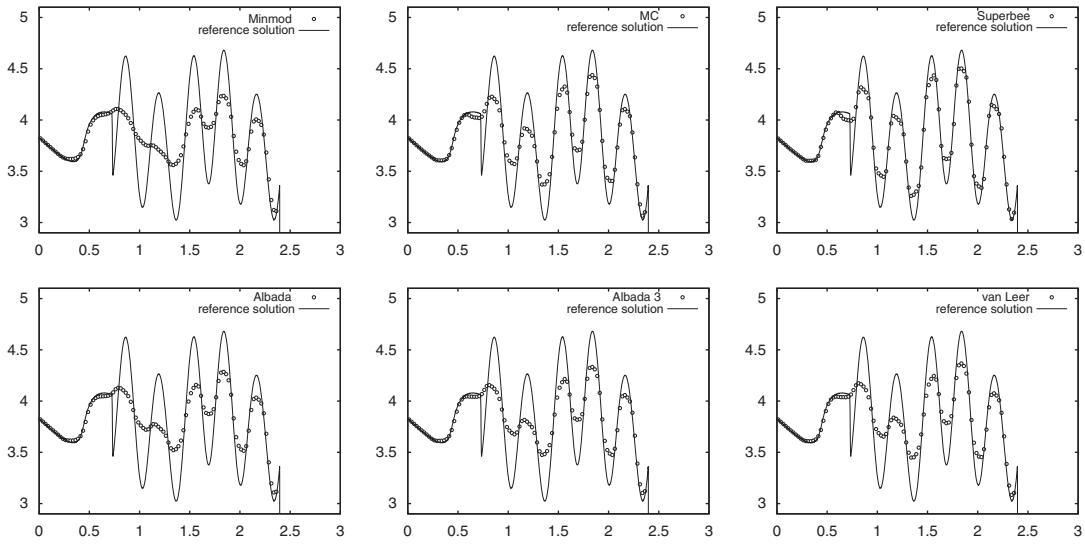


Figure 26. Shu–Osher problem at $t = 1.8$; results for CFL-independent limiters: Minmod, MC, Superbee (top row); Albada, Albada 3, van Leer (bottom row); resolution: 400 cells.

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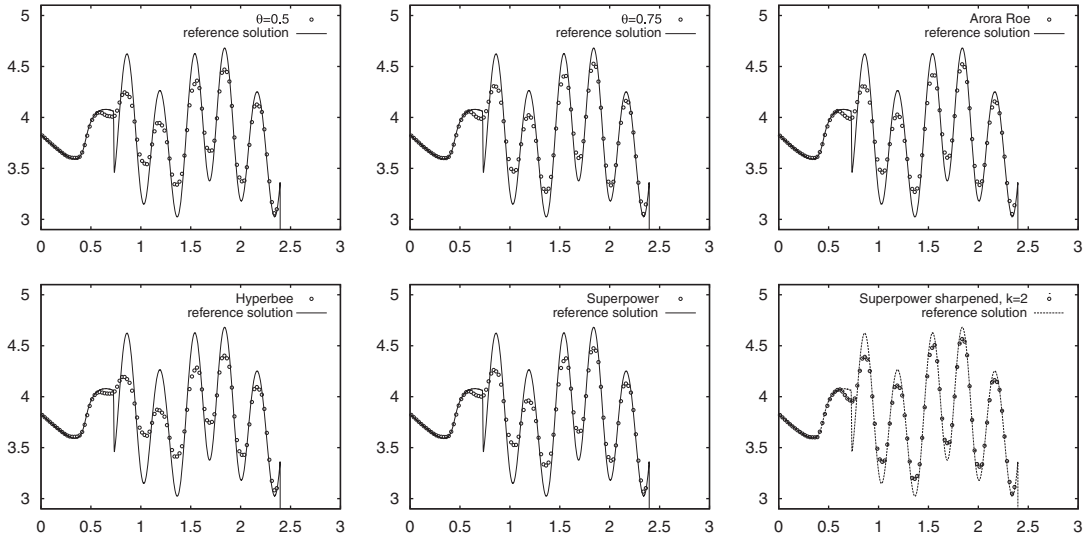


Figure 27. Shu–Osher problem at $t=1.8$; results for third-order adapted limiters: θ-versions of Arora–Roe with $\theta=0.5$, $\theta=0.75$, and $\theta=1.0$ (top row); Hyperbee and Superpower (bottom row); resolution: 400 cells.

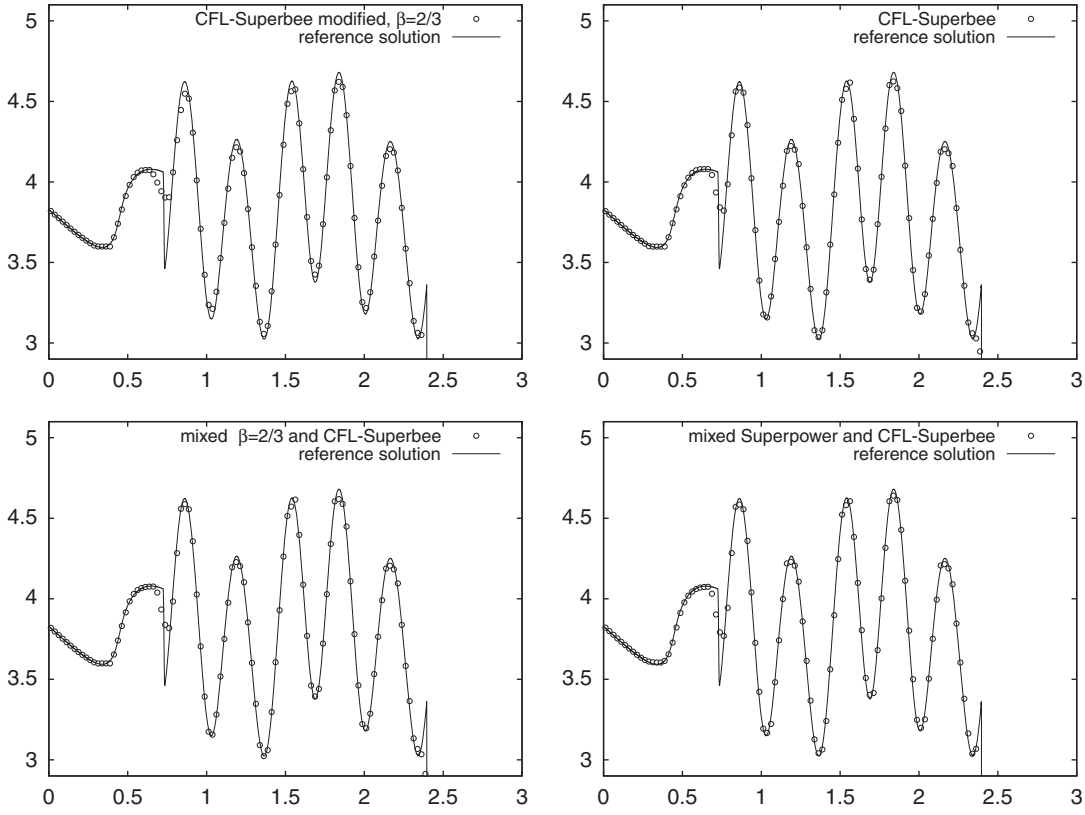


Figure 28. Shu–Osher problem at $t=1.8$; results for Superbee-type limiters: modified CFL–Superbee with $\beta=\frac{2}{3}$ and original CFL-Superbee (top row); CFL-Superbee for linear wave and modification with $\beta=\frac{2}{3}$ for non-linear waves (left) and Superpower for non-linear waves (right) (bottom row); resolution: 400 cells.

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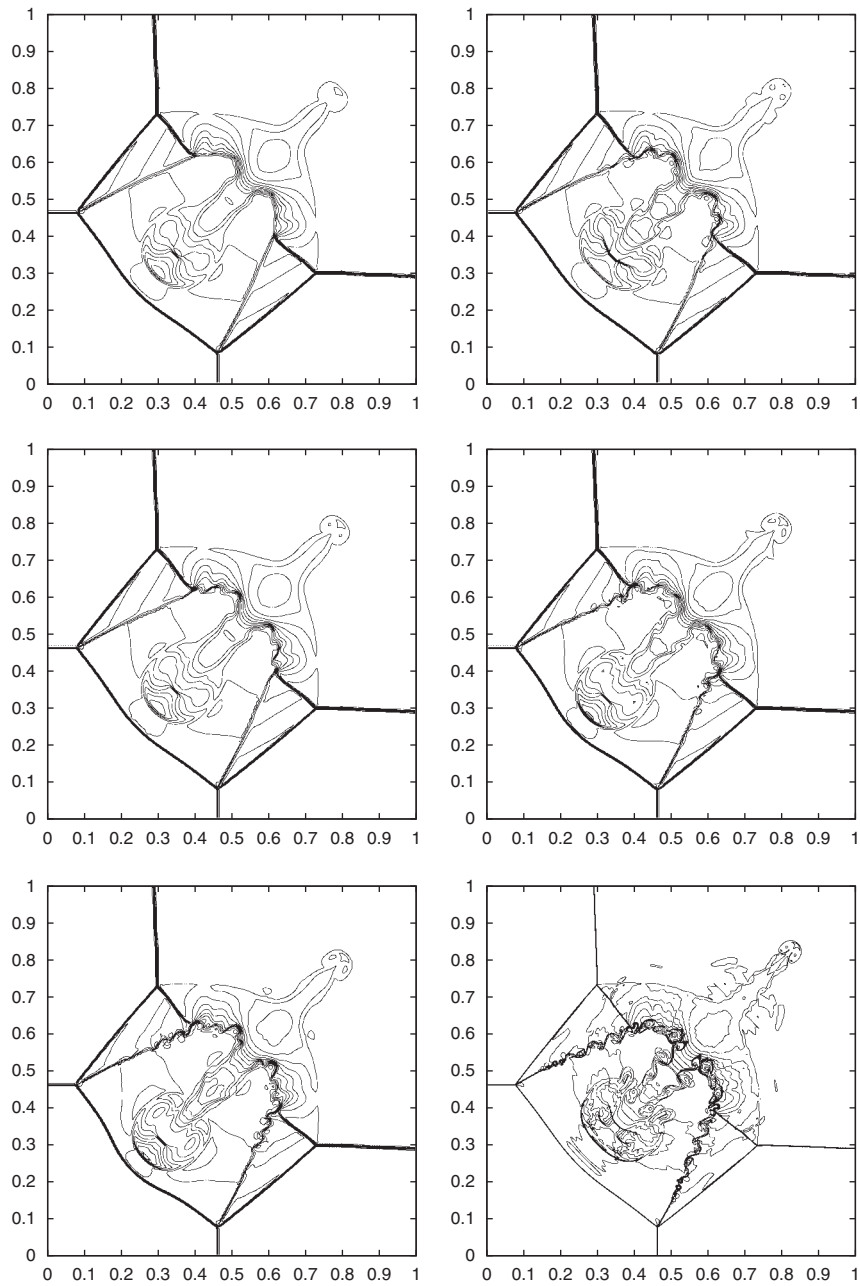


Figure 29. Two-dimensional Riemann problem at $t = 0.8$; results for MC and Superbee (top row); Superpower on non-linear waves and on linear waves: Arora–Roe and the modified Superbee with $\beta = \frac{2}{3}$ (middle row); CFL-Superbee and reference solution (bottom row); resolution: 200×200 cells, reference solution 800×800.

limiter yields better results than its CFL-independent counterpart. The $\beta = \frac{2}{3}$-limiter gives almost the same resolution as CFL-Superbee, but reduces the slight overshoot at the rightmost wave.

5.5. The Gresho vortex

As a third example for the 2d Euler equations, we present the Gresho vortex. In contrast to the previous examples, it is subsonic and, therefore, does not include shocks. The Gresho vortex is transported without any change. The centrifugal force is balanced by a suitable pressure distribution.

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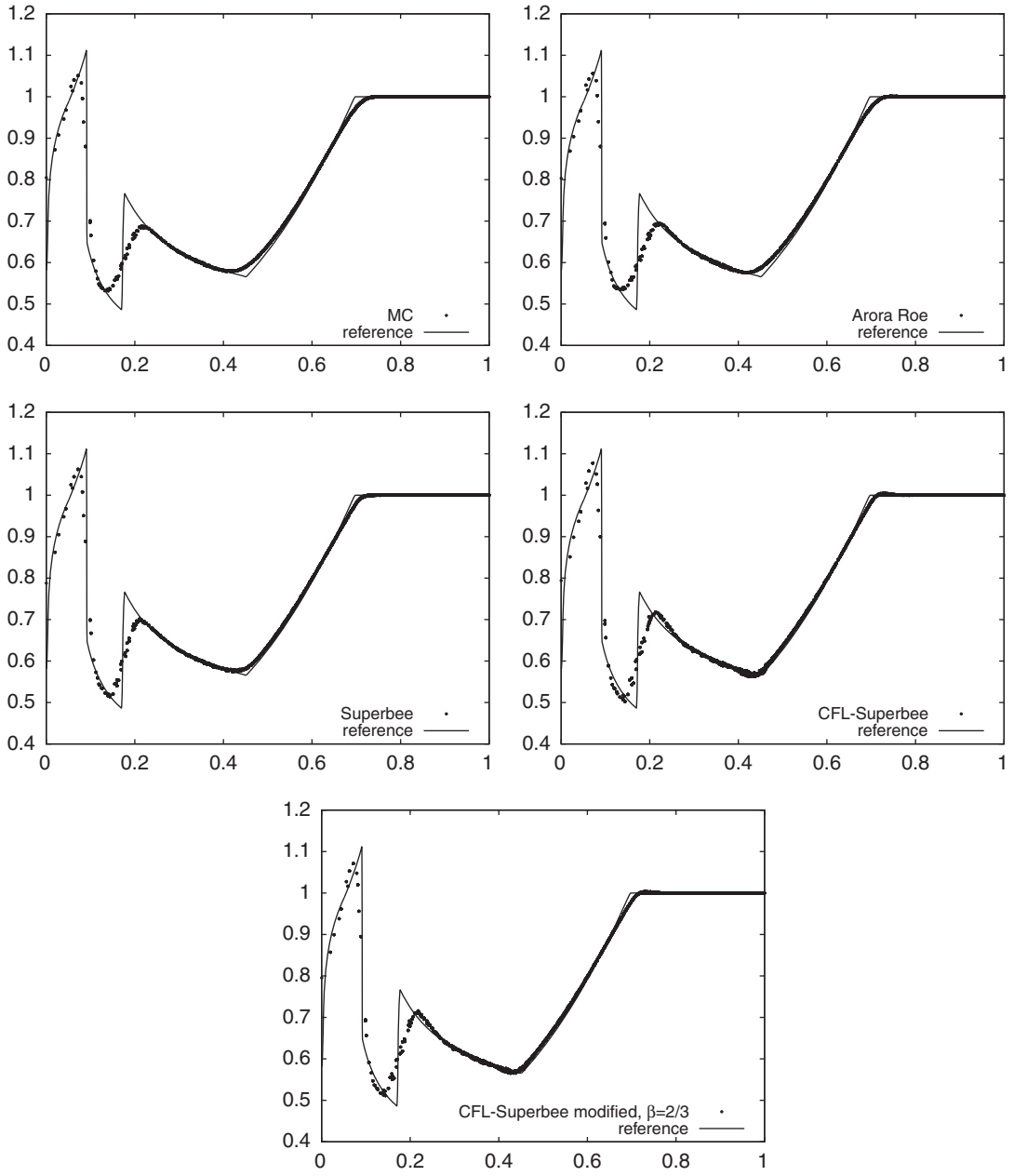


Figure 30. Scatter plots for implosion problem at $t=0.25$. Top row: MC and Arora–Roe; middle row: Superbee and CFL-Superbee; bottom row: $\beta=\frac{2}{3}$-limiter.

We employ the settings used by Liska and Wendroff [10]. The angular velocity of the vortex at rest depends only on the radius:

$$u_\phi(r)=\begin{cases}5r, & 0\leqslant r<0.2;\\ 2-5r, & 0.2\leqslant r<0.4;\\ 0, & 0.4\leqslant r.\end{cases} \tag{80}$$

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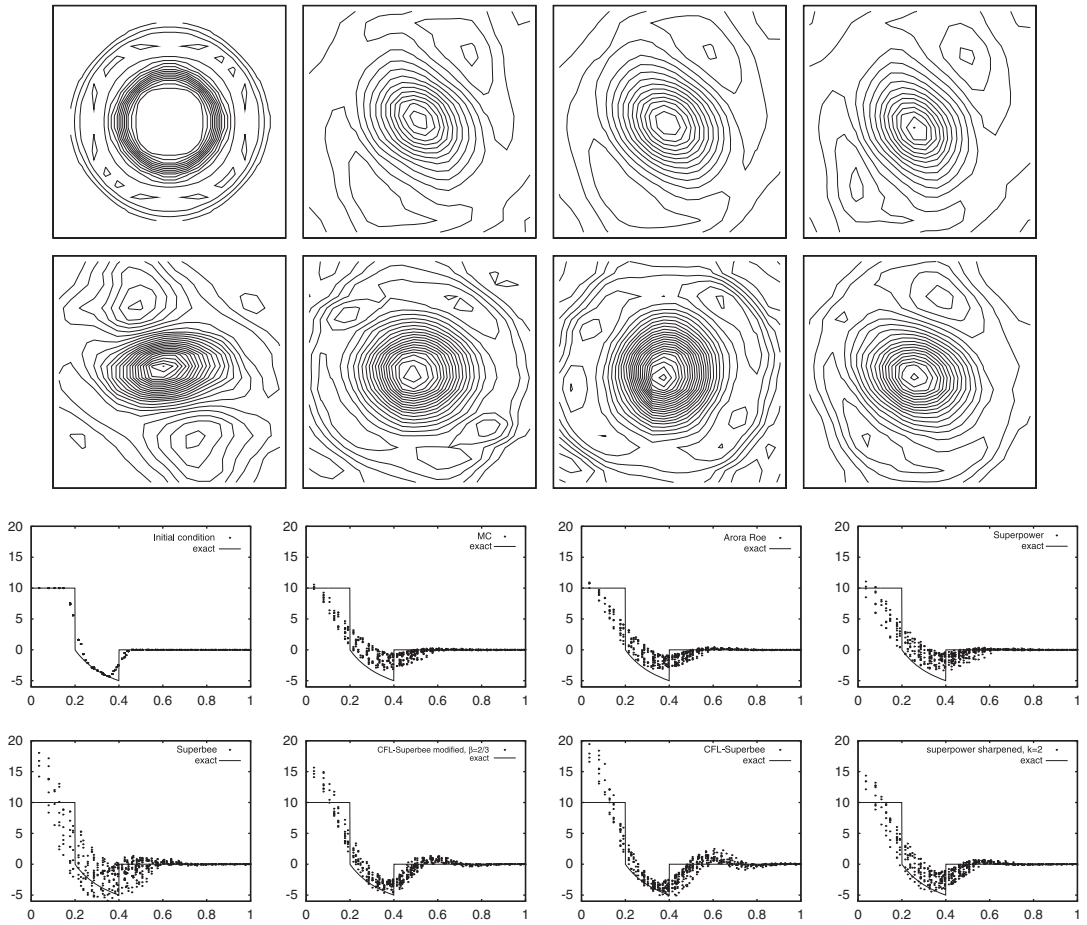


Figure 31. Vorticity for moving Gresho vortex on 80×20 grid at $t=3$, contours and scatter plots (over radius). Top row: exact solution, MC, Arora–Roe, Superpower. Bottom row: Superbee, $\beta=\frac{2}{3}$, CFL-Superbee, sharpened Superpower with $k=2$.

Consequently, the pressure distribution also only depends on the radius:

$$u_\phi(r)=\begin{cases}5+\frac{25}{2}r^2, & 0\leqslant r<0.2;\\ 9-4\ln 0.2+\frac{25}{2}r^2-20r+4\ln r, & 0.2\leqslant r<0.4;\\ 3+4\ln 2, & 0.4\leqslant r.\end{cases} \tag{81}$$

For the moving vortex, we superimpose the resulting velocity field by an advection with velocity 1. The computations are done for the rectangle $[0,4]\times[0,1]$, the initial position of the vortex center is $(0.5,0.5)$. In the y-direction, we impose transmissive boundary conditions, in the x-direction periodic boundaries. Figures 31 and 32 show results on a 80×20 and a 160×40 grid at time $t=3$. They show that besides CFL-Superbee and the $\beta=\frac{2}{3}$ version, all other limiters lead to a significant loss in symmetry. Superbee produces the worst results. The less compressive limiters, whose results are presented in the upper row, suffer from their diffusive character, whereas the CFL-Superbee, especially on the coarser grid, highly overestimates the vorticity near the center of the vortex. The best overall performance is found with our newly introduced third-order adapted CFL-Superbee with $\beta=\frac{2}{3}$.

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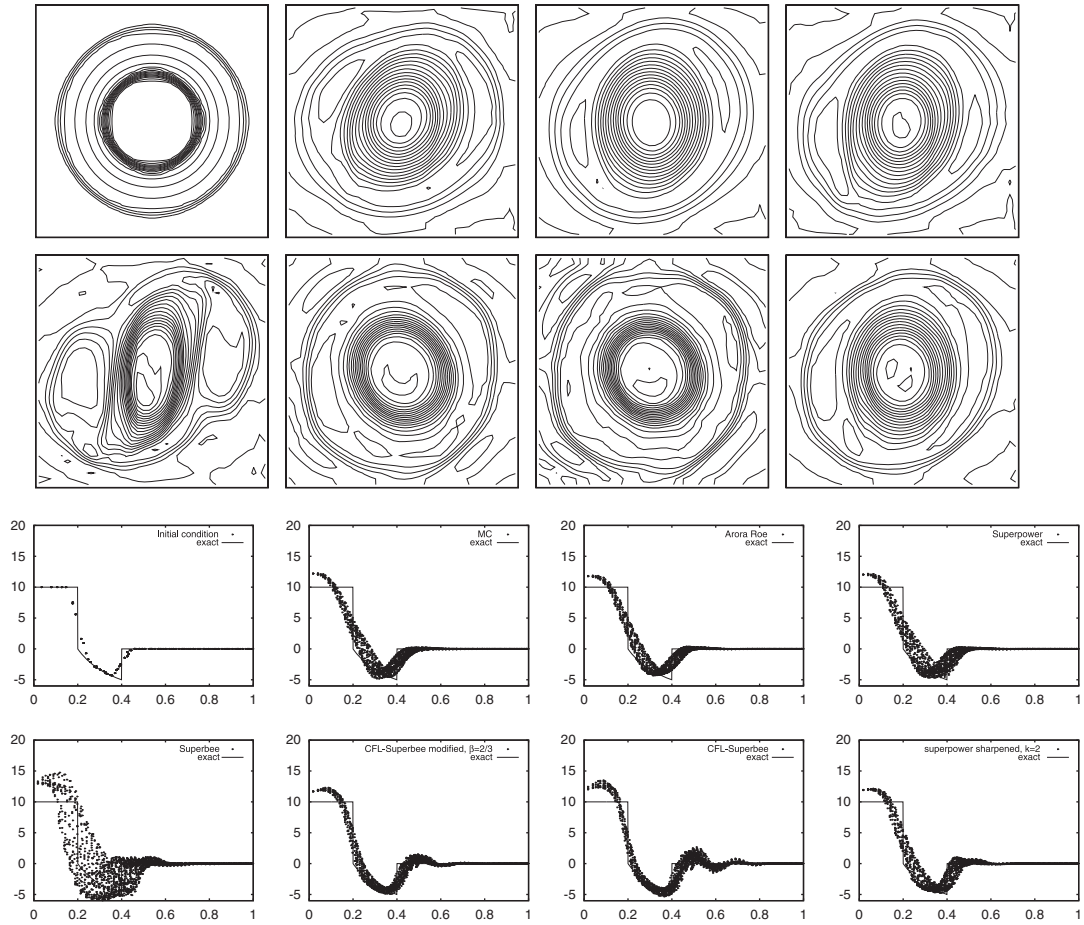


Figure 32. Vorticity for moving Gresho vortex on 160×40 grid at $t=3$, contours and scatter plots (over radius). Top row: exact solution, MC, Arora–Roe, Superpower. Bottom row: Superbee, $\beta=\frac{2}{3}$, CFL-Superbee, sharpened Superpower with $k=2$.

6. CONCLUSIONS

TVD-schemes still play an important role in the simulation of compressible gas flow and other advection-dominated problems. The crucial part of the scheme is the limiter used. Formally the schemes are of second order. According to the theory of Roe, reviewed in Section 2.2, they approximate the amplitudes with fourth order but the phase, and, thus, the shape of the waves, only with second order. Our results show that, as was already mentioned by Arora and Roe [2], a scheme that is in some sense adapted to the third-order linear scheme in the vicinity of linear states, which means that the ratio r is close to one, shows much of the properties of a third-order scheme. As a consequence, the shape of waves is reproduced with a higher order of accuracy. We showed that the old limiters are not to be considered as fixed schemes but rather as models for large families of limiters. We introduced some new modifications, such as the adaption of the Superbee-limiter to the third-order scheme, by introducing a new parameter β. Furthermore, we presented a modification of the Arora–Roe limiter to allow for the reconstruction of local extrema without violating the TVD-condition. We also applied the theory of Section 2 to enhance the modern non-TVD-limiter given by Čada and Torrilhon. Based on an idea of Serna and Marquina, we constructed a CFL-number-dependent improved version of the van Leer limiter, the Superpower limiter, which proved very useful in the limiting of non-linear waves. And finally, we introduced a more compressive but still everywhere differentiable version of the van Albada limiter.

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The main results of the numerical tests are the following: For piecewise linear data and linear waves, it is preferable to employ CFL-Superbee. The standard Superbee would, by its higher numerical viscosity, destroy the piecewise linear shapes and, by the squaring effect, make everything into rectangular waves. If the data are not piecewise linear, the $\beta = \frac{2}{3}$ version of CFL-Superbee is the best choice. The loss in the amplitudes is minimal, whereas the squaring effect is considerably reduced. For non-linear waves, a less compressive limiter should be used. Best choices are, according to the results in Section 5.2, the θ-versions of the Arora–Roe limiter or, even better, but at a slightly higher computational cost, a smooth limiter like Superpower.

By a suitable choice of limiters, one can obtain results that compete with ENO and WENO schemes of third or fifth order [22, 32], at least at standard grid resolutions.

Since the limiters under consideration are designed for 1d use or, in two or three dimensions, direction-wise limiting, there is still some work to be done to transfer the high quality of CFL-dependent limiters to the framework of unstructured grids.

Where to find the limiters. The computations are done with *clawpack* [17]. The files needed to use the limiters with *clawpack* can be found on the authors website www.math.uni-sb.de/ag/kemm/codes or www.math.tu-cottbus.de/kemm/codes. Hence, readers can do their own numerical experiments or use the implemented limiters as models to develop their own specialized versions.

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